Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

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Working draft — v0.8

Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances degenerate on large graphs (von Luxburg, Radl, and Hein). We show the geometry does not vanish: it is largely carried by the *angular* coordinate. Writing each node's low-mode embedding in polar form—magnitude times direction—the direction preserves graph geodesics (Spearman $\rho \approx 0.93$ on a reference torus), while the magnitude, and an equal-size random-mode representation, preserve almost none. Keeping only the direction is the row-normalization of Ng–Jordan–Weiss spectral clustering applied to the commute-weighted embedding; its existing theory (Schiebinger, Wainwright, and Yu) concerns cluster separation, and we supply a metric-geometric account. An exact identity locates the mechanism in *tangential noncollapse*—the embedding velocity must not turn purely radial—rather than in radius constancy; from it we prove a deterministic angular-transfer theorem, a conditional graph-to-manifold result (carried by the random-walk eigenvectors and the sup-norm eigenvector rates of Calder, García Trillos, and Lewicka), and an *unconditional* bi-Lipschitz statement on the flat torus. The radial claim is stated at three levels: proved for the full embedding at intrinsic dimension $d \geq 3$, empirical for the truncated radius the algorithm computes, and borderline at $d = 2$. Uniformity of the metric constants in the mode count fails for the commute filter, but the *rank* form is uniform below dimension four: for $d \leq 3$ the angular ranking converges to a fixed Green-kernel ranking, exact on the circle and on two-point homogeneous spaces. The extended empirical program—substrate robustness across graph families, a manifold diagnostic, and the fidelity–dimension trend—is the companion Paper I.b.

Key words. spectral embedding, commute-time (resistance) distance, row normalization, spectral clustering, geodesic distance, intrinsic dimension, Laplacian eigenmaps
AMS subject classifications. 05C50, 58J50, 62H30, 05C81, 68T09

1 Introduction

Spectral embeddings—Laplacian eigenmaps [3], diffusion and commute-time maps [9]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [26]. Yet the most successful spectral-clustering algorithm [17] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [23, 25]. It does not address metric geometry. Closer to our object, unit-hypersphere normalization has been applied *directly* to the commute-time embedding for 3D shape registration across differing samplings [24]; there it is a sampling-robustness heuristic, and the geodesic content of the resulting angular distances is not analyzed. **This paper asks what row-normalization recovers of the graph's geodesic structure, and why.**

Our answer is a clean decomposition. Write each node's low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is the truncated, normalized counterpart of the radial coordinate that von Luxburg's theorem kills for the *full, unnormalized* commute embedding at $d \geq 3$ (Corollary 1); empirically it tracks local degree (§5.2). The *direction* carries the graph's geodesic geometry (after the α=1 density normalization where sampling is strongly non-uniform, Paper I.b). Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

The underlying operation—*separate scale from direction, then keep the part that carries the task geometry*—recurs beyond clustering: polar-coordinate KV-cache quantization splits key vectors the same way [14], evidence for the *decomposition*, not for always discarding the radius. Our hypotheses fix the boundary: discarding the magnitude is the right quotient precisely when the radial variable is nuisance for the metric preserved—the tangential-noncollapse condition (A2) of Theorem 7—which for graph geodesics holds, the radius being degree/density noise (Corollary 1). On the reference torus the direction recovers geodesics where the magnitude and a random-mode representation do not (§5); the companion Paper I.b [1] establishes the basis's substrate-robustness and its failure modes across graph families.

Contributions. The first is the **Keep-the-Angle Principle** (§3), which comprises: a Radial-Degeneracy corollary for the full embedding (Corollary 1, from [26]), with truncation controlled by $1/\lambda_{m+1}$ (Proposition 2); an exact angular-distortion identity raised to a deterministic angular-transfer theorem for arbitrary filtered embeddings (Theorem 7, Corollary 8), from which Angular Preservation follows conditionally on the graph (Theorem 9, Appendix B) and unconditionally on the flat torus and round sphere (Corollaries 11, 12); a distortion-to-rank bridge (Proposition 13); and the filter dichotomy, with a uniform lower angular bound and a Green-kernel rank limit for $d \leq 3$ (Theorem 25). The second is a confirming measurement on the torus—controls (§5), theorem verification (§5.2), and a weighting ablation (§5.3); the extended empirical program is the companion Paper I.b [1]. We do not claim to derive physics: the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [3] and diffusion maps [9] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [26] proved the commute distance of a large geometric graph degenerates, $R(i,j) \to 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. They also propose an *amplified* commute distance as a correction, the natural competitor to ours; rather than repair the radius we discard it and keep the angle. **Row-normalization.** Ng–Jordan–Weiss [17] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [23] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k-means' post-normalization success; consistency of spectral clustering is treated in [25].

Table 1: Status of the paper's claims at a glance.

Claim	Status	Where
Radial degeneracy (full commute)	proved (vL–R–H + bdd. deg.)	Cor. 1
Truncated normalized radius $\sim$ degree	empirical	§5.2
Truncation controlled by $1/\lambda_{m+1}$	proved	Prop. 2
Exact angular-distortion identity	proved	Prop. 6
Deterministic transfer (A1)–(A3)	proved	Thm. 7
Graph transfer at fixed m	proved, cond. (H1)–(H2)	Thm. 9
Global form	cond. (H3)	Cor. 10
Flat torus, $m = 4$	proved, uncond.	Cor. 11
Round sphere, $m = d + 1$	proved, uncond.	Cor. 12
Distortion $\Rightarrow$ rank	proved	Prop. 13
Heat filter, uniform in m	proved (continuum)	Thm. 14
Commute *upper*, fixed-scale unif.	refuted (exact $\mathbb{S}^1$)	Prop. 15
Commute *lower* bound, unif. in m (tori)	proved	Prop. 22
Rank limit = Green-kernel ranking ($d \le 3$)	proved	Thm. 25
Filter phase diagram, critical $d = 4$	proved	Prop. 23
Plain-filter rank collapse on $\mathbb{S}^1$	proved	Thm. 17
Angular preservation (rank form)	conjecture; empirical	Conj. 3
Unnormalized differential noncollapse	proved (monotone)	Prop. 4

Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. The same normalization removes degree as a nuisance in degree-corrected block models—SCORE [15] cancels it through eigenvector ratios, and spherical spectral clustering under the DCBM [20] projects to the sphere—but those are community-recovery guarantees, not geodesic ones. The embedding Ψ itself is Rustamov's Global Point Signature [21] from shape analysis, and the magnitude we discard is used in its own right for cluster and anomaly detection [8]; we keep the direction and give it a metric reading. **Hypergraph-rewriting graph families.** Hypergraph rewriting [27, 12] generates large graphs from a local rule; we use several such rules as one of five substrate families, treated purely as graph generators (any physical interpretation is deferred to §6). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [13, 22, 18]; we test and reject a hyperbolic reframe in Paper I.b.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k), $0 = \lambda_0 < \lambda_1 \le \cdots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = \big(v_1(i)/\sqrt{\lambda_1}, \ldots, v_m(i)/\sqrt{\lambda_m}\big)$, the eigenvectors unit-normalized ($\sum_i v_k(i)^2 = 1$, i.e. $VV^\top = I$; the empirical convention $\frac{1}{n}\sum_i v_k(i)^2 = 1$ rescales the numerical truncation bounds by n but leaves all angular statements, which are scale-invariant, unchanged). Write $\Psi_i = r_i\,\hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$. Throughout, "angular distance" means the Euclidean *chordal* distance between unit vectors, $\|\hat{u}_i - \hat{u}_j\| = 2\sin(\theta_{ij}/2)$ with θ_{ij} the angle between $\hat{u}_i$ and $\hat{u}_j$; it is monotone in θ_{ij}, so all rank (Spearman) statements are unchanged by the chordal-vs-arc choice, and the figures use the chordal form. [1]

[1] We call Ψ "commute-style" rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n-1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance $R(i,j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. Dimension regimes use this measured intrinsic dimension, with one convention enforced throughout: the radial-degeneracy *theorem* is gated at $d \geq 3$; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and is treated empirically (§5.2); and in (quasi-)1D the radius stays geodesic-informative, as predicted (Paper I.b). The angular *effect* is observed from quasi-1D upward.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 3$ satisfying the conditions of [26] (the $d = 2$ case is marginal: at the recurrent boundary the resistance carries logarithmic corrections, matching the borderline behaviour we measure in §5.2), and let Φ_i be the* full *commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L^+_{ii}$. Assume additionally that the degree ratio $\max_i k_i / \min_i k_i$ stays bounded (as it does with high probability in the graph models of [26]). As $n \to \infty$ the resistance distance degenerates uniformly and multiplicatively, $\sup_{i \neq j} \left| R(i,j)/(1/k_i + 1/k_j) - 1 \right| \to 0$ [26], and this pairwise statement transfers to the diagonal: $L^+_{ii} = (1/k_i)(1+o(1))$ uniformly over nodes, so the radius satisfies $\|\Phi_i\| = (1/\sqrt{k_i})\,(1+o(1))$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

The diagonal transfer is not automatic: the pairwise limits determine only combinations $L^+_{ii} + L^+_{jj} - 2L^+_{ij}$, and a gauge ($\sum_j L^+_{ij} = 0$) plus a trace identity, with the degree-ratio bound controlling error propagation, close the gap (Appendix A).

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting needs (i) the normalized analogue of the degeneracy [26] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 3). Numerically the truncated radius tracks the local degree, $\rho(r_i, 1/\sqrt{k_i}) = 0.79$–$0.87$ across $m = 5$–40, while the full normalized radius is near-constant (coefficient of variation 0.09) and radius-only geodesic preservation is $\rho \leq 0.08$ (§5.2).

Proposition 2 (Inverse-next-eigenvalue truncation bound). *Let $\Psi^\infty_i = \left(v_k(i)/\sqrt{\lambda_k}\right)_{k=1}^{n-1}$ be the full symmetric-normalized commute-style embedding and Ψ^m_i its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi^\infty_i\|^2 - \|\Psi^m_i\|^2 \leq \frac{1}{\lambda_{m+1}}, \qquad 0 \leq \|\Psi^\infty_i - \Psi^\infty_j\|^2 - \|\Psi^m_i - \Psi^m_j\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda^{-1}_{m+1} \sum_{k>m} v_k(i)^2$ and $\lambda^{-1}_{m+1} \sum_{k>m} (v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}_{k=0}^{n-1}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for $i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m}(v_k(i) - v_k(j))^2 \leq \sum_k (v_k(i) - v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2\sum_k v_k(i)v_k(j) = 2$. These identity sums run over all $k = 0, \ldots, n-1$, including the trivial mode $k = 0$ that Ψ omits; it sits only in the subtracted low-mode partial sums, and dropping its nonnegative terms can only tighten the tail bounds. $\square$

Two consequences. First, the error is controlled by the inverse spectral gap $1/\lambda_{m+1}$ alone. This is *not* an n-independent absolute bound: on a fixed-degree neighbourhood graph refined toward the continuum λ_{m+1} itself shrinks, so $1/\lambda_{m+1}$ grows. What it controls is the truncation error *relative to the embedding scale*, which is fixed by the same low eigenvalues; since the angular and rank statistics we report are invariant to global scale, that relative control is what transfers, and is why the continuum refinement of Paper I.b does not degrade. Second, the deleted content is high-frequency (large-λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5). But this bound is a *sanity check, not the angular theorem*: it controls the full-vs-truncated discrepancy at a fixed graph and mode cutoff, in a norm weighted by inverse eigenvalues; it does not show the deleted tail is irrelevant to angular *rank* geometry, and it gives no uniform-in-n angular stability statement. The empirical flatness in m (§5.2) remains *evidence for* Conjecture 3, not a consequence of Proposition 2. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 2 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 3.

Conjecture 3 (Angular Preservation). *For ε-graph or k-NN samples of a compact Riemannian manifold in the density class of [11, 6], row-normalization $\hat{u}_i = \Psi_i/\|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise angular distances remain* rank-faithful *to geodesic distance—Spearman $\geq \rho_0 > 0$—uniformly over the admissible mode schedule: m multiplet-closed at a spectral gap with $m \geq m_0(M)$ (the immersion threshold of [2, 19]); sampled pairs above the resolution scale θ_n; and, for growing $m = m(n)$, the spectral-window condition $\mathcal{E}_n(\mu_m)\gamma_m^{-1} \to 0$ of Appendix B. Under strongly non-uniform sampling the claim is asserted for the Coifman–Lafon α=1 angle (Paper I.b); empirically $\rho_0 \approx 0.9$.*

We reserve "rank-faithful" for the limit $\liminf_n \rho_S \geq \rho_0 > 0$ along admissible schedules $(n, m(n))$, with $\rho_0 = \rho_0(M)$ per manifold (the Green-rank coefficient of Theorem 25 for $d \leq 3$); the diagnostic reading—using the small-world/dimension-spread screen of Paper I.b to decide whether a *given* graph lies in the class—is a separate corollary-in-use, kept out of the conjecture to avoid circularity. "Uniformly" means a lower bound over the admissible family, not over all integers m: too few modes need not embed M, a truncation inside a multiplet is basis-dependent, and m may not outrun the sample.

A stronger *metric* form—bi-Lipschitz to geodesic distance with constants uniform in m and n at fixed scale—is now known **false** for the commute weighting: high modes raise the local angular speed like $\sqrt{\Lambda}$ (Proposition 15, exact on S^1), so the metric strengthening must be either scale-dependent (uniform above the wavelength $\Lambda^{-1/2}$, open) or filtered (for the heat weighting it is a continuum theorem, Theorem 14). The rank clause is untouched: the empirical flatness in m is rank stability at a fixed mode budget, and our evidence is rank-based.

Metric evidence on the torus. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale the angular map has a robust empirical bi-Lipschitz constant $\kappa = 2.66 \pm 0.03$ (ratio of the 97.5th to 2.5th percentiles of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs, five independent draws), against 3.15 ± 0.04 for the full commute embedding, 7.1 ± 0.3 for an equal-size random-mode basis, and 128 ± 4 for the magnitude; its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.064 ± 0.016 (random modes: 0.985 ± 0.004); the map stays injective—nothing is folded together. This is metric distortion, not rank correlation—one substrate, but a well-understood one: the correct uniform metric target is scale-dependent for the commute filter (Proposition 15).

The uniform-in-m frontier. The metric form of Conjecture 3 rests on one named hypothesis, which we isolate rather than assume away.

(H1)-uniform. *At mode-counts m taken at a spectral gap, the truncated eigenmap $F = (\varphi_k/\sqrt{\mu_k})_{k\le m}$ is L_0-bi-Lipschitz onto its image on geodesic r_0-balls, with L_0 and r_0 independent of m.*

What is known is weaker: [5] and the quantitative truncated-eigenmap embeddings of [2, 19] give, for each fixed m past an immersion threshold, *some* constant $L_0(m)$, and as m grows the added eigenfunctions oscillate on wavelength $\mu_m^{-1/2}$, so holding L_0 fixed asks that no mode ever create a near-fold—the degeneration a symmetry or thin neck can force. Two endpoints bound the question: on the flat torus the *two-sided* bound holds at the first multiplet (Corollary 11, the Clifford embedding) and the *lower* half holds uniformly in m (Proposition 22), and the property is *filter-specific*—a theorem for the heat weighting (Theorem 14), with the upper half false at fixed scale for the commute weighting (Proposition 15). The empirical flatness in m (§5.2) supports the rank form; the metric form reduces to one clean question in spectral geometry:

Proposition 4 (The unnormalized differential does not collapse)**.** *For nested multiplet-closed cutoffs $m < m'$ and every tangent v, $\|dF_{m'}(v)\|^2 = \|dF_m(v)\|^2 + \sum_{m<k\le m'} |d\varphi_k(v)|^2/\mu_k \ge \|dF_m(v)\|^2$, so $\sigma_{\min}(dF_m(x))$ is nondecreasing in m; once F_{m_0} is an immersion (finite m_0 [5]), compactness gives $\inf_{m\ge m_0} \inf_x \sigma_{\min}(dF_m(x)) \ge c > 0$.*

The *unnormalized* lower bound is therefore automatic; the obstruction to two-sided uniformity lives entirely in the normalization—the angular lower bound $\|N_m(x) - N_m(y)\| \ge c\, d_M(x, y)$ must survive a denominator $\|F_m\| \to \infty$, which it does on flat tori (Proposition 22) and is conjectured in general (Conjecture 3), while the *upper* constant diverges for commute (Proposition 15). What remains for the *metric* form of Conjecture 3 is the normalized lower bound beyond the torus, closed at fixed m by Theorem 9 and Lemma 20. The *rank* form, however, is settled below the critical dimension: for $d \le 3$ the commute angular ranking converges uniformly in m to the Green-kernel ranking (Theorem 25), reducing Angular Preservation to positivity of a geometric Green-rank coefficient—equal to 1 on the circle and on two-point homogeneous spaces of dimension ≤ 3 (Corollary 26).
A proof—or a counterexample forced by symmetry—is a self-contained problem we do not attempt; none of the paper's proved results (Table 1) depends on its resolution.

3.1 Toward a proof of Conjecture 3

Proposition 2 controls truncation; the remaining question is whether row-normalization preserves the geometry the low modes carry. It does, and the mechanism is an exact identity. To set expectations plainly: the payoff of this subsection is a *transfer* theorem, not a proof of the conjecture—it reduces the conjecture, at fixed m, to two continuum hypotheses (local eigenmap bi-Lipschitzness, H1, and radius-flat transversality, H2), plus an injectivity hypothesis (H3) for the global form.

Lemma 5 (Normalization preserves bi-Lipschitz geometry)**.** *For nonzero $u, v \in \mathbb{R}^m$ with $\hat{u} = u/\|u\|$,*

$$\|\hat{u} - \hat{v}\|^2 = \frac{\|u - v\|^2 - \big(\|u\| - \|v\|\big)^2}{\|u\|\,\|v\|}.$$

If $F : (X, d) \to \mathbb{R}^m$ satisfies $L^{-1}d(x, y) \le \|F(x) - F(y)\| \le L\, d(x, y)$ with $\rho^- \le \|F\| \le \rho^+$ and radius Λ-Lipschitz, then $\hat{F} = F/\|F\|$ obeys $\|\hat{F}(x) - \hat{F}(y)\| \le (L/\rho^-)\, d(x, y)$, and if $\Lambda < 1/L$,

$$\|\hat{F}(x) - \hat{F}(y)\| \;\ge\; \frac{\sqrt{L^{-2} - \Lambda^2}}{\rho^+}\, d(x, y).$$

Proof. The identity combines $\|\hat{u}-\hat{v}\|^2 = 2-2\langle u, v\rangle/(\|u\|\|v\|)$ with $\langle u, v\rangle = \frac{1}{2}(\|u\|^2+\|v\|^2-\|u-v\|^2)$. The upper bound uses numerator $\le \|u - v\|^2 \le L^2d^2$ and $\|u\|\|v\| \ge (\rho^-)^2$. For the lower bound, $(\|u\| - \|v\|)^2 \le \Lambda^2d^2$ and $\|u - v\|^2 \ge d^2/L^2$ give numerator $\ge d^2(L^{-2} - \Lambda^2)$; divide by $\|u\|\|v\| \le (\rho^+)^2$. □

The condition $\Lambda < 1/L$ is a quantitative radial degeneracy: the radius must vary slower than the embedding moves, and as $\Lambda \to 0$ (the limit of Corollary 1) normalization becomes a pure scaling with lower constant $1/(L\rho^+)$. On the 2-torus the identity is exact to 10^{-15}, but the margin is razor-thin ($\Lambda = 0.0918$ against lower stretch $A = 0.0920$), so the lemma's worst-case constant is near-vacuous there—while the pointwise tangential component never degenerates (0/9200 local pairs radially degenerate; measured local angular constant 2.2). This survives on a *non-homogeneous* manifold, where the radius genuinely varies and a radial fold could in principle occur: on the swiss-roll (a 2D manifold carried into $\mathbb{R}^{12}$, §4) only 1/29,481 local pairs across five draws are radially degenerate, the local angular bi-Lipschitz constant is finite (3.67 ± 0.40), and the radius Lipschitz constant sits well below the lower stretch ($\Lambda_{\text{med}}/A = 0.30 \pm 0.02$)—so (A2) holds off the homogeneous case too. Proposition 6 explains why, and Theorem 7 makes the resulting two-point tangential noncollapse (A2) the hypothesis itself, with $\Lambda < 1/L$ only one sufficient route to it—so the thin radius margin costs the theory nothing.

Proposition 6 (Exact angular distortion)**.** *Let F be differentiable with $r = \|F\| > 0$ and $\hat{F} = F/r$. Taking $y \to x$ in Lemma 5 gives, for every tangent vector v,*

$$\|d\hat{F}(v)\|^2 = \frac{\|dF(v)\|^2 - dr(v)^2}{r^2} = \frac{\|dF(v) - \hat{u}\, dr(v)\|^2}{r^2}, \qquad dr(v) = \langle dF(v), \hat{u}\rangle.$$

The angular stretch in a direction v is *exactly* the tangential part of $dF(v)$—the component orthogonal to the radius—divided by r. So $\hat{F}$ is a local bi-Lipschitz immersion *iff* dF is nowhere purely radial ($dF(v) \not\parallel F$ for $v \neq 0$), the lower stretch being $\inf_v \|dF(v) - \hat{u}\, dr(v)\|/(r\|v\|)$. This sharpens Lemma 5: the crude $\Lambda < 1/L$ merely forces the radial term $dr(v)$ below the total $\|dF(v)\|$, whereas the proposition shows what is really required is that the embedding's velocity never aligns with its position. Lemma 5 and Proposition 6 link the two halves of the principle and reduce the conjecture to standard ingredients.

The general transfer principle. Proposition 6 identifies the mechanism infinitesimally; its finite-distance form can be made the *hypothesis* itself. The result is a transfer theorem that knows nothing about manifolds, Laplacians, or eigenvectors—and of which the eigenmap theorem below (Theorem 9) is the commute-weighted special case. Any uniformly calibrated discrete embedding inherits local angular bi-Lipschitz geometry from a continuum embedding whenever the continuum displacement has a two-point tangential component bounded below; radius flatness is only one sufficient route to that tangential noncollapse.

Theorem 7 (Deterministic angular transfer)**.** *Let (X, d) be a metric space, $\{x_1, \dots, x_n\} \subset X$, and $F : X \to \mathbb{R}^m \setminus \{0\}$ with radius $R = \|F\|$ and angular map $N = F/R$. Let $Z_{n,i} \in \mathbb{R}^m \setminus \{0\}$ be* any *discrete embedding of x_i, with angular rows $\hat{Z}_{n,i} = Z_{n,i}/\|Z_{n,i}\|$, and fix a pair domain $\mathcal{P} \subset X \times X$*

(e.g. $\{0 < d(x,y) \le r_0\}$). Assume: ***(A1)*** *$0 < \rho^- \le R \le \rho^+ < \infty$ on the relevant region;* ***(A2) two-point tangential noncollapse:*** *there are $0 < \tau \le \beta < \infty$ with*

$$\tau\, d(x,y) \;\le\; \Big(\|F(x)-F(y)\|^2 - \big(R(x)-R(y)\big)^2\Big)^{1/2} \;\le\; \beta\, d(x,y) \qquad \textit{for all } (x,y) \in \mathcal{P}$$

(the middle quantity is exactly the part of the displacement that survives row-normalization; infinitesimally it is the tangential component of Proposition 6); ***(A3) uniform calibration:*** *there are a global scale $s_n > 0$, an orthogonal $Q_n \in O(m)$, and $E_n \to 0$ with $\max_{i\le n}\|Q_n(s_n Z_{n,i}) - F(x_i)\| \le E_n$. Then for every $\eta \in (0,1)$, once $E_n < \rho^-/2$, every sampled pair in $\mathcal{P}$ with*

$$d(x_i,x_j) \;\ge\; \theta_n(\eta) := \frac{4E_n\,\rho^+}{\eta\,\tau\,\rho^-} \;\longrightarrow\; 0$$

obeys

$$(1-\eta)\,\frac{\tau}{\rho^+}\,d(x_i,x_j) \;\le\; \|\hat{Z}_{n,i} - \hat{Z}_{n,j}\| \;\le\; \Big(\frac{\beta}{\rho^-} + \eta\,\frac{\tau}{\rho^+}\Big)\,d(x_i,x_j).$$

Proof. The identity of Lemma 5 with $u = F(x)$, $v = F(y)$ reads $\|N(x)-N(y)\|^2 = \big[\|F(x)-F(y)\|^2 - (R(x)-R(y))^2\big]/\big(R(x)R(y)\big)$, so (A1)–(A2) make N bi-Lipschitz on $\mathcal{P}$ with constants $a = \tau/\rho^+$ and $b = \beta/\rho^-$. Pairwise angular distances of $\hat{Z}$ are invariant under the scale s_n and the orthogonal Q_n, and by (A3) with normalization stability (Lemma 19, using $\|F(x_i)\| \ge \rho^-$ in the denominator) each calibrated angular row lies within $2E_n/\rho^-$ of $N(x_i)$; hence each pairwise angular distance lies within $4E_n/\rho^-$ of the continuum one. For $d(x_i,x_j) \ge \theta_n(\eta)$ that error is at most $\eta a\, d(x_i,x_j)$; adding and subtracting it from the continuum bounds gives the claim. □

Corollary 8 (Filtered spectral embeddings)**.** *Let M and m (at a spectral gap) be as in Theorem 9, and let h be a positive Lipschitz spectral filter on the retained window ($H_m := \max_{k\le m} h(\mu_k)$, L_h a Lipschitz constant). Define the filtered eigenmap $F_h(x) = \big(h(\mu_k)\varphi_k(x)\big)_{k=1}^m$ and graph rows $Z_{n,i}^{(h)} = \big(h(\tilde\lambda_k)\, v_k(i)\big)_{k=1}^m$ with $\tilde\lambda_k = \lambda_k/c_n$. Under the sup-norm spectral consistency of Appendix B (Lemma 20 with $\mathrm{diag}(\mu_k^{-1/2})$ replaced by $\mathrm{diag}(h(\mu_k))$; same proof), hypothesis (A3) holds with high probability with*

$$E_{n,h} \;=\; H_m\,\epsilon_n \;+\; L_h\,\delta_n\,\sqrt{m}\,(\Phi_m + \epsilon_n) \;\longrightarrow\; 0.$$

Hence if F_h satisfies (A1)–(A2) on $\{d_M \le r_0\}$, then with high probability the angular map of the h-filtered graph embedding is bi-Lipschitz to d_M, with the constants of Theorem 7, for all sampled pairs with $\theta_n(\eta) \le d_M \le r_0$. The commute filter $h(\lambda) = \lambda^{-1/2}$ recovers Theorem 9; diffusion $h(\lambda) = e^{-t\lambda}$ and plain $h \equiv 1$ are equally admissible at fixed *m. The weighting ablation of §5.3—plain eigenmaps collapse as m grows while commute and diffusion stay flat—is therefore a statement about how each filter's constants $(\tau, \beta, \rho^\pm)$ behave* in *m, not about the transfer mechanism, which is filter-independent.*

Theorem 9 (Conditional angular transfer for the commute-weighted eigenmap; local form of Conjecture 3)**.** *Let M be a compact d-manifold whose graph Laplacian spectrally converges to M's ([11] for ε-graphs, [6] for k-NN graphs), and fix a mode-count m* at a spectral gap *of M. Assume* ***(H1)*** *the truncated eigenmap $F : x \mapsto (\varphi_k(x)/\sqrt{\mu_k})_{k=1}^m$ is L_0-bi-Lipschitz onto its image on geodesic balls of radius r_0—the quantitative truncated-eigenmap embedding property of [5, 2, 19]—and* ***(H2)*** *the associated truncated radius is Λ_0-Lipschitz with $\Lambda_0 < 1/L_0$ and lies in $[\rho^-, \rho^+]$. Set*

$$A = \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{2\rho^+}, \qquad B = \frac{2L_0}{\rho^-},$$

both independent of n. Then there is a resolution scale $\theta_n \to 0$ (explicitly, a constant multiple of the sup-norm spectral-consistency rate of [6]) such that, with high probability as $n \to \infty$, the graph angular map obeys $A\, d_M(x,y) \le \|\hat{u}_x - \hat{u}_y\| \le B\, d_M(x,y)$ for all sampled x, y with $\theta_n \le d_M(x,y) \le r_0$.

Proof (outline; complete proof in Appendix B). This is the special case of Theorem 7 and Corollary 8 with $h(\lambda) = \lambda^{-1/2}$ and $\eta = 1/2$: (H1)–(H2) imply the tangential noncollapse (A2) with $\tau = \sqrt{L_0^{-2} - \Lambda_0^2}$ (positive because $\Lambda_0 < 1/L_0$) and $\beta = L_0$, giving $A = (1-\eta)\tau/\rho^+$ and $B \ge \beta/\rho^- + \eta\tau/\rho^+$. Note (H2) is only one sufficient route: the theorem survives with (H2) replaced by (A2) directly, which is strictly weaker and is the condition the torus actually satisfies with a healthy margin (the discussion after Lemma 5). Self-contained proof: by (H1), (H2) and Lemma 5, the *continuum* angular map $N = F/\|F\|$ is bi-Lipschitz on r_0-balls with constants $a = \sqrt{L_0^{-2} - \Lambda_0^2}/\rho^+$ and $b = L_0/\rho^-$ (Lemma 18). The transfer to the graph needs *sup-norm* eigenvector convergence—the ∞-norm rates of [6] rather than generic L^2 consistency—so that normalization survives where $\|F\|$ is smallest. It is applied to the *random-walk* eigenvectors, to which the graph angular map reduces *exactly* by per-node degree cancellation ($\Psi_i = \sqrt{k_i}\, \Psi_i^{\mathrm{rw}}$; Step 0 (iv) of Appendix B), so the transfer needs no degree-concentration hypothesis and holds under non-uniform sampling. After the graph-to-continuum eigenvalue scaling and a block-orthogonal alignment across the multiplets below the gap (both invisible to pairwise angular distances), the graph embedding is calibrated to F within an error $E_n \to 0$ *uniformly over nodes* (Lemma 20); in particular the graph radius inherits $[\rho^- - E_n, \rho^+ + E_n]$, and the graph angle lands within $2E_n/\rho^-$ of N (Lemma 19). The two-sided bound then follows with $A = a/2$ and $B = b + a/2 \le 2L_0/\rho^-$ for every sampled pair above the scale $\theta_n = 8E_n/(a\rho^-)$. The cutoff is necessary, not an artifact: below the graph's resolution, combinatorially twin nodes receive identical entries in every eigenvector below the localized twin-difference mode—in particular in every low mode the angular map uses—so $\hat{u}_i = \hat{u}_j$ while $d_M > 0$ (Appendix B). □

From local to global. Theorem 9 is local because (H1) is. One further hypothesis—the global sibling of the radial-fold transversality of Proposition 6—globalizes it.

Corollary 10 (Global conditional form)**.** *Assume (H1)–(H2) and additionally **(H3)**: the continuum angular map $N = F/\|F\|$ is injective on M. Let $c_0 := \min\{\|N(x) - N(y)\| : d_M(x,y) \ge r_0\}$, which is positive by (H3) and compactness. Then, with*

$$A_g = \min\Big(A,\ \frac{c_0}{2\,\mathrm{diam}\,M}\Big), \qquad B_g = \max\Big(B,\ \frac{2}{r_0}\Big),$$

with high probability as $n \to \infty$ the graph angular map obeys $A_g\, d_M(x,y) \le \|\hat{u}_x - \hat{u}_y\| \le B_g\, d_M(x,y)$ for all *sampled x, y with $d_M(x,y) \ge \theta_n$ (proof: Appendix B, Step 5).*

(H3) is genuinely extra: Portegies-style mode counts [19] make F itself injective, but $N(x) = N(y)$ requires only that F *radially align* two points ($F(x) = tF(y)$, $t > 0$), which injectivity of F does not exclude. Empirically it holds on the torus—the angular map folds nothing together (metric-evidence paragraph above). On the flat torus, in fact, nothing at all is left to assume:

Corollary 11 (Unconditional instance: the flat torus)**.** *Let $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with $m = 4$ (the full first multiplet, a spectral gap). Then (H1)–(H3)* hold, *with explicit constants: the truncated eigenmap is the Clifford embedding $F(x,y) = \sqrt{2}\,(\cos x, \sin x, \cos y, \sin y)$, which is L_0-bi-Lipschitz* globally *($r_0 = \mathrm{diam}\,M$) with $L_0 = \sqrt{2}$; the radius is constant, $R \equiv 2$, so $\Lambda_0 = 0$ and $\rho^\pm = 2$; and*

$N = F/2$ is injective. Consequently, with high probability as $n \to \infty$, the graph angular map of an ε-graph or k-NN sample of the flat torus is globally bi-Lipschitz to geodesic distance above the resolution scale, unconditionally, *with $A = 1/(4\sqrt{2})$ and $B = \sqrt{2}$; the sharp continuum angular distortion is exactly $\pi/2$ (the chord-vs-arc ratio $2\sin(\delta/2)/\delta \in [2/\pi, 1]$ per coordinate—the source of the near-antipodal dip in the measured κ). The statement is basis-independent: any L^2-orthonormal basis of the four-dimensional eigenspace is an $O(4)$ rotation of the Clifford one, invisible to angular distances (proof: Appendix B, Step 6).*

The measured graph-level $\kappa \approx 2.6$ (at m=10, which *splits* a multiplet; the gap-closed cutoffs on $\mathbb{T}^2$ are $4, 8, 12, 20, \ldots$) sits above the $\pi/2 \approx 1.57$ continuum floor by finite-n and truncation slack; m=4 is the theoretically clean budget. The transfer theorem certifies only $B/A = 8$, so the measured $\kappa \approx 2.6$ sits between the sharp continuum $\pi/2$ and the certified 8—a gauge of how loose the transfer constants are.

Corollary 12 (Unconditional instance: the round sphere)**.** *On $S^d \subset \mathbb{R}^{d+1}$ with $m = d+1$ (the ℓ=1 eigenspace, eigenvalue d), $F \propto x$: the radius is constant and $N = F/\|F\| = x$ is the identity embedding, so N is globally bi-Lipschitz with the sharp constants $[2/\pi, 1]$ ($\|N(x) - N(y)\| = 2\sin(\theta/2)$, $\theta = d_{S^d}(x, y)$), unconditionally—a second constant-curvature instance ($\mathbb{T}^d$ likewise extends to $m = 2d$).*

More generally, on any compact homogeneous space transitivity makes the radius constant at every multiplet-closed cutoff, so (H2) holds automatically and the metric conjecture reduces there to (H1) and (H3) alone.

From metric to rank. The conjecture's currency is rank (Spearman); the theorems' is distortion. The bridge is elementary and deterministic.

Proposition 13 (Distortion controls rank)**.** *Let $\mathcal{P}$ be a finite set of node pairs on which $A' d_M \le \widehat{D} \le B' d_M$ holds for some embedded distance $\widehat{D}$, write $\kappa = B'/A'$, and let $\hat{\mu}(\kappa)$ be the fraction of unordered pairs $\{P, Q\} \subset \mathcal{P}$ whose geodesic distances lie within ratio κ of each other ($\max / \min < \kappa$). Then, absent ties (ties, e.g. from integer hop distances, enter only through the standard tie corrections), Kendall's $\tau(\widehat{D}, d_M) \ge 1 - 2\hat{\mu}(\kappa)$ and Spearman's $\rho_S \ge 1 - 3\hat{\mu}(\kappa)$ up to an $O(1/|\mathcal{P}|)$ finite-sample correction.*

Proof. A discordant pair-of-pairs ($d_P < d_Q$ but $\widehat{D}_P > \widehat{D}_Q$) forces $A' d_Q \le \widehat{D}_Q < \widehat{D}_P \le B' d_P$, i.e. $d_Q/d_P < \kappa$; so the discordant fraction is at most $\hat{\mu}(\kappa)$ and $\tau = 1 - 2\,(\text{discordant fraction}) \ge 1 - 2\hat{\mu}(\kappa)$. Daniels' inequality $|3\tau - 2\rho_S| \le 1$ [10] converts this to $\rho_S \ge (3\tau - 1)/2 \ge 1 - 3\hat{\mu}(\kappa)$. □

Combined with Corollary 10, this reduces the rank clause of Conjecture 3, at fixed m, to (H1)–(H3) *plus* a distance-ratio condition on the substrate: $\rho_0 = 1 - 3\mu(\kappa) > 0$ whenever $\mu(\kappa) < 1/3$ (sub-θ_n pairs are a vanishing fraction and enter the bound as $o(1)$). The reduction is honest about its weakness: it is worst-case over *all* κ-distorted maps. For the small-ball distance law $f(r) \propto r^{d-1}$ on $[0, r_0]$ an exact computation (Appendix B, Step 7) gives $\mu(\kappa) = 1 - \kappa^{-d}$, so the certificate $\tau \ge 2\kappa^{-d} - 1$ is non-vacuous only for $\kappa < 2^{1/d}$: a fixed distortion budget certifies *exponentially* less rank structure as the intrinsic dimension grows, and on substrates whose pairwise distances concentrate—the high-d regime—no finite distortion certifies any rank fidelity at all. This is a genuine ceiling for distortion-based guarantees, and a second, rank-specific mechanism for the fidelity-vs-dimension decay of Paper I.b. That the measured decay is *linear* ($-0.157\,d$, Paper I.b), far milder than the worst case, says the eigenmap angle is far from the worst κ-distorted map: the empirical ≈ 0.9 levels remain stronger than what distortion alone implies, which is why that clause of the conjecture stays empirical.

Uniformity in the truncation: what is true, and what is false. Everything so far is at fixed m. The uniformity-in-m question splits by filter, and both halves can now be answered in the continuum. Throughout, the cutoff is spectral and multiplet-closed: the map retains all eigenspaces with $\mu_k \le \Lambda$, so $m = m(\Lambda)$, matching the spectral-gap convention.

Theorem 14 (Uniform-in-truncation angular embedding: heat filter)**.** *Let M be compact, smooth, without boundary, and for $t > 0$ let $F_{t,m}(x) = \big(e^{-t\mu_k/2}\varphi_k(x)\big)_{k=1}^{m}$ be the heat-filtered eigenmap, with $N_{t,m} = F_{t,m}/\|F_{t,m}\|$. There is $t_0(M) > 0$ such that for every fixed $t \le t_0$ there exist $m_0(t)$, $r_t > 0$, and $0 < a_t \le b_t < \infty$ with*

$$a_t\, d_M(x,y) \;\le\; \|N_{t,m}(x) - N_{t,m}(y)\| \;\le\; b_t\, d_M(x,y) \qquad \text{whenever } d_M(x,y) \le r_t,$$

for every *$m \ge m_0(t)$: the constants do not deteriorate as $m \to \infty$. (Proof: Appendix C, by heat-kernel diagonal asymptotics and C^2 tail control; the exponential filter is what makes the tail summable against the polynomial growth of eigenfunction norms.)*

Proposition 15 (Commute cutoff: tangential noncollapse, scale-dependent metric)**.** *Let $F_\Lambda(x) = \big(\mu_k^{-1/2}\varphi_k(x)\big)_{0<\mu_k\le\Lambda}$ be the commute-weighted spectral cutoff, $R_\Lambda = \|F_\Lambda\|$, $N_\Lambda = F_\Lambda/R_\Lambda$. There are $c > 0$ and Λ_0 such that for all $\Lambda \ge \Lambda_0$, uniformly over $x \in M$ and unit tangent v,*

$$\big(\|dF_\Lambda(v)\|^2 - dR_\Lambda(v)^2\big)^{1/2} \;\ge\; c\,\Lambda^{d/4}:$$

the tangential component never collapses—it grows *with the cutoff. Meanwhile $R_\Lambda \asymp \Lambda^{d/4-1/2}$ for $d > 2$ (and $\asymp \sqrt{\log\Lambda}$ for $d = 2$), so the angular derivative grows, $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$ ($d > 2$; $\sqrt{\Lambda/\log\Lambda}$, $d = 2$). Consequently no m-independent fixed-scale upper Lipschitz constant exists for the commute angular map: its natural resolution scale is the spectral wavelength $\Lambda^{-1/2}$. On S^1 this is exact: with all modes to frequency N, R_N is constant, $\|dF_N\|^2 = N$, and $\|dN_N\| \asymp \sqrt{N}$. (Proof: Appendix C, via the differentiated pointwise Weyl law of Lemma 21; the S^1 case needs no Weyl input and is exact.)*

The two statements split the uniformity question cleanly. For the *heat/diffusion* filter—one of the two weightings the ablation of §5.3 finds flat in m—uniform-in-truncation local bi-Lipschitzness is a continuum theorem. For the *commute* filter the tangential mechanism survives uniformly, with room to spare, but the metric constants are intrinsically scale-dependent: high modes raise the local angular speed like $\sqrt{\Lambda}$, so the correct uniform metric object is bi-Lipschitzness *above the wavelength scale* $\Lambda^{-1/2}$— equivalently, a mode budget matched to the scale of interest, which is the mode-budget phenomenon of Paper I.b handed a mechanism. The empirical "flat in m" of §5.2 is thereby correctly read as *rank* stability under a fixed mode budget, not as an m-independent fixed-scale metric theorem. One gap remains at this point: the $\sqrt{\Lambda}$ angular-speed growth of Proposition 15 is filter-generic (the plain filter has the same order), so it explains the budget phenomenon but not the ablation's dramatic *rank* separation between plain and commute (§5.3). The next results address that gap: on a general manifold the plain filter *concentrates* (Proposition 16), and on the circle—the canonical homogeneous instance—its angular ranking provably collapses (Theorem 17).

Proposition 16 (Plain-filter distance concentration)**.** *Let $F^{(1)}_\Lambda(x) = \big(\varphi_k(x)\big)_{0<\mu_k\le\Lambda}$ be the unweighted (plain-eigenmap) cutoff with angular map $N^{(1)}_\Lambda$, and fix $\delta > 0$. Then, uniformly over pairs with $d_M(x,y) \ge \delta$,*

$$\big\|N^{(1)}_\Lambda(x) - N^{(1)}_\Lambda(y)\big\|^2 \;=\; 2 - \frac{2\,E_\Lambda(x,y)}{\sqrt{E_\Lambda(x,x)\,E_\Lambda(y,y)}} \;=\; 2 + O\big(\Lambda^{-1/2}\big):$$

all separated plain angular distances concentrate at the orthogonality value $\sqrt{2}$, and the vanishing correction is carried by the off-diagonal spectral function $E_\Lambda(x, y)$—an oscillatory kernel with sign changes on the wavelength scale $\Lambda^{-1/2}$ and no monotone relation to d_M. (Proof: Appendix C.)

Concentration alone does *not* imply that the ordering degrades: Spearman and Kendall correlations are invariant under a shrinking positive-affine rescaling, so a correction of the form $\varepsilon_\Lambda f(d_M)$ with $\varepsilon_\Lambda \to 0$ could preserve the rank order for every Λ. What the *oscillatory* form of E_Λ precludes is precisely that residual monotonicity. We make this exact on the canonical homogeneous instance, where E_Λ is the Dirichlet kernel and the collapse can be computed in closed form. For the *commute* filter the same expansion instead gives correction kernel $G_\Lambda(x, y) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)\varphi_k(y)$, low-frequency dominated (for $d \le 3$ it converges off the diagonal to the Green's function)—a coherent, distance-structured kernel; on $\mathbb{S}^1$ it is the monotone periodic Green's function, so the commute ordering there is preserved.

Theorem 17 (Exact plain-filter rank collapse on $\mathbb{S}^1$)**.** *On the circle $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$, let the plain eigenmap retain the first N complete Fourier multiplets, $F_N(x) = (\cos x, \sin x, \ldots, \cos Nx, \sin Nx) \in \mathbb{R}^{2N}$, with angular map $\widehat{F}_N = F_N/\|F_N\|$ (so $\|F_N\|^2 \equiv N$). Let X, Y be independent uniform points, $\Theta = d_{\mathbb{S}^1}(X, Y) \sim \mathrm{Unif}[0, \pi]$, and $Q_N(\Theta) = \|\widehat{F}_N(X) - \widehat{F}_N(Y)\|^2$. Then the population Spearman and Kendall correlations against geodesic distance both vanish:*

$$\rho_S\big(\Theta, Q_N(\Theta)\big) \xrightarrow[N\to\infty]{} 0, \qquad \tau_K\big(\Theta, Q_N(\Theta)\big) \xrightarrow[N\to\infty]{} 0.$$

Thus on $\mathbb{S}^1$ the plain angular ordering retains asymptotically no geodesic information, even though every separated distance concentrates at $\sqrt{2}$ (Proposition 16). (Proof: Appendix C.)

The mechanism in one line: the $1/\mu$ commute weighting is a low-pass filter on the *correction* kernel that carries the ranks, while equal weighting hands the correction to the oscillatory top of the spectral window—which is why the ablation separates the filters at the rank level (0.06 vs 0.89 at m=20, §5.3) even though their crude metric orders coincide. For graph transfer at growing $m = m(n)$, the calibration of Lemma 20 extends whenever $\mathcal{E}_n(\mu_m)\,\gamma_m^{-1} \to 0$ (γ_m the gap isolating the retained block; $\mathcal{E}_n$ the graph-consistency error, polynomially growing in μ_m), and a $C^{0,1}$ eigenvector theory [7] is the route to shrink the resolution cutoff toward the connectivity scale (supplement, Appendix C).

What remains is delimited. **(H1)** is subtle: Jones–Maggioni–Schul [16] guarantee bi-Lipschitz charts only from a well-chosen d-subset of eigenfunctions per ball, and low-spectrum multiplicity makes a truncation inside a multiplet basis-dependent—so m is taken at a spectral gap (the m=4 of Corollary 11) or the map phrased through spectral projectors. **(H2)** is, by Proposition 6, sharper than radius flatness: the eigenmap's velocity must be nowhere purely radial, with (A2) its two-point form and the hypothesis to verify in practice; it holds on the tested manifolds, homogeneous and not (§3), and whether the eigenmap image ever develops a radial fold on some manifold is the sharp remaining question. **(H3)** is genuinely extra: injectivity of F [19] does not exclude a radial alignment $F(x) = tF(y)$, so the global statement is carried by (H3) (Corollary 10)—while on the torus (H1) holds globally and Corollary 11 needs none of the chart hypotheses (H1)–(H3) (the sampling model and spectral consistency of Theorem 9 are still assumed).
Status. Table 1 records what is proved, conditional, empirical, and open; at fixed m the rank clause reduces to (H1)–(H3) plus a distance-ratio condition on the substrate (Corollary 10, Proposition 13), and uniformity in m is settled in the continuum by filter. The honest gap: Corollary 1 explains why the full unnormalized radius *dies* ($d \ge 3$), not why the angle *lives*—angular fidelity persists ($\rho \approx$ 0.88–0.90) even on quasi-1D graphs where the corollary does not apply, so the angular half is carried

by the eigenmap-embedding mechanism. That mechanism is analyzable: if the low eigenvectors converge to Laplace–Beltrami eigenfunctions [4, 11] and eigenfunction coordinates furnish locally bi-Lipschitz charts [16], geodesic fidelity should follow from the chart geometry—the natural route to Conjecture 3, whose residual difficulty is isolated to the normalization by Proposition 4. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i—and, under strongly non-uniform sampling, keep the α=1 density-normalized angle (Paper I.b).

Dimension gating. Corollary 1 is a $d \geq 3$ theorem; $d = 2$ is the recurrent borderline, where the degeneracy appears with logarithmic corrections and we treat it empirically (§5.2). In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.2.

4 Methods

Table 2: Protocol by section. n: nodes; m: retained modes; n_a: anchors; D: independent graph draws reported (mean±sd when $D > 1$; single-draw rows marked † per the headline-uncertainty policy below).

Section	substrate(s)	graph	n	m	n_a	D
§5 core	2-torus	kNN	2000	5–80	150	$1^\dagger$ (5 at m=10)
§5.2	2-torus	kNN	2000–4000	5–80	200	5 (m=10)
§5.3	2-, 3-torus	kNN	2000/3000	10–40	150	$1^\dagger$ (10 ctrl)

Where a manifold ground truth exists we report both the measured consensus dimension d_G and the true d_M (2-torus d_M=2; 3-torus 3; swiss-roll 2; king-lattice 2). On the controls the estimators read a few percent low (a known finite-sample curvature bias): the consensus $d_G \approx 1.94$ on the 2-torus, with the individual estimators 5–8% below d_M at n=2000. All dimension-indexed claims use d_G consistently: estimator bias shifts the axis, not the ordering of the Paper I.b trend.

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread ⇒ clean manifold, and the agreed value is our measured intrinsic dimension d. **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (chordal distances between unit-normalized low-mode embeddings, §3), *random* (m random modes; control), *magnitude* (radius only), and *Isomap* (classical MDS of the graph geodesics, an oracle given the geodesics). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ for the substrate benchmarks, 200 for the torus metric tests (§3.1)—19,900 pairs). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers), but pairs share anchor nodes and are not independent, so we treat pair-resampling CIs as optimistic floors; the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, m=10: 0.914 ± 0.009 over five draws). The reference torus therefore appears at slightly different angle-ρ across tables (0.89–0.93) as mode-count, anchor count, and graph draw vary by section; each table uses its section's protocol (m=20, n_{anchor}=150 unless stated). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding samples. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v), u the roll angle and v the height—carried by a random orthogonal map into $\mathbb{R}^{12}$ with isotropic noise (σ=0.03) and connected by a k=10 nearest-neighbour graph on n=2000 ambient points. It stands in for sequential/embedding-derived data (each item a point on a low-d

semantic manifold, consecutive items nearby), not a specific text corpus. The Lorentzian causal set is an n=2000 Poisson sprinkling in a 1+1D Minkowski diamond (light-cone coordinates u, v uniform on the unit square); $p \prec q$ iff $u_p < u_q$ and $v_p < v_q$, and each point is linked to its k=6 nearest causal-*past* neighbours in proper time $\tau^2 = \Delta t^2 - \Delta x^2$, symmetrized for the undirected geodesic BFS—so edges encode short causal *relations* (nearest past neighbours in proper time, which need not be Hasse links), and the intrinsic d=2 is the sprinkling dimension.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3. On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag.×dir.)	**angle only**	magnitude only
4	0.820	**0.928**	0.024
8	0.703	**0.933**	0.027
12	0.595	**0.900**	0.030
20	0.525	**0.891**	0.019

The cutoffs are multiplet-closed (the gap-closed budgets $4, 8, 12, 20$ on $\mathbb{T}^2$), and each entry is a mean over five independent graph draws (angle standard deviation ≤ 0.009; e.g. m=20 gives 0.891 ± 0.009). The angle carries the geometry (≈ 0.9, flat across the admissible schedule); the magnitude carries little geodesic information in these benchmarks; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2); averaged over ten random draws $\rho = 0.08 \pm 0.16$, individual draws occasionally catching a low mode and spiking to ~ 0.4.

5.2 Verifying Corollary 1 and Conjecture 3

We check the two claims directly. The angle-ρ is flat in graph size while the full commute distance decays, and the truncated radius tracks degree ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.79$–$0.87$ across $m = 5$–40; radius-only $\rho \leq 0.08$). For the full *unnormalized* commute radius the degeneracy is exact, Corollary 1 holding cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \to 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 3: in an m-sweep at $n = 4000$, angle-ρ is flat ≈ 0.93 across $m = 5$–80 while the full embedding decays $0.773 \to 0.437$.

5.3 Which weighting carries the angle?

The direction depends on the per-mode weight, so we ablate it on the torus (d=2) and a 3-torus (d=3). Plain eigenmaps (w_k=1) give an angle that *collapses* with mode count—angle-ρ falls from 0.554 ± 0.013 at m=10 to 0.081 ± 0.012 at m=20 (five independent draws, d=2)—because equal weighting lets high modes swamp the direction. The commute weight $1/\sqrt{\lambda_k}$ is flat instead ($0.918 \pm 0.006 \to 0.896 \pm 0.006$), as is diffusion $e^{-\lambda_k t}$; both suppress high modes, which is what the geometry needs. The "flat in m" property is thus a property of the weighting, not of angles in general—and it selects the commute (or diffusion) weight, not the plain eigenmap. Proposition 16 now supplies the continuum mechanism: plain angular distances concentrate at $\sqrt{2}$ with only an oscillatory, geometry-blind correction, while the commute correction is the (low-pass, Green-type) kernel that

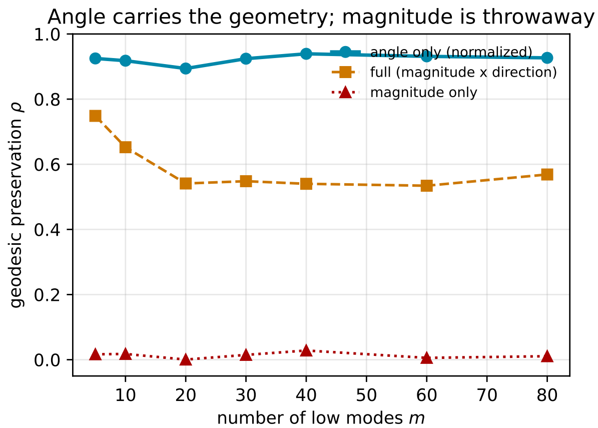

Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) versus mode-count m. Angle carries the geometry; magnitude carries little geodesic signal; the full commute embedding decays with m.

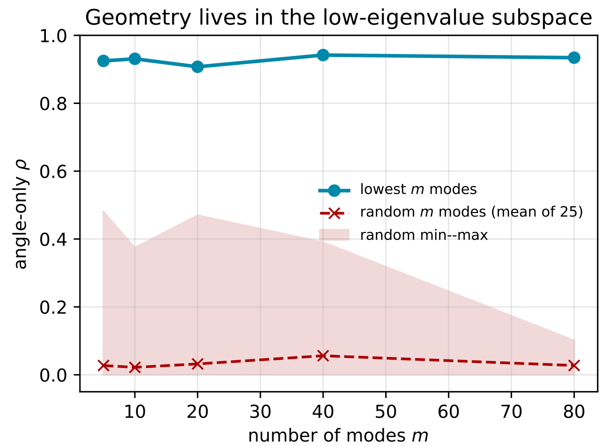

Figure 2: Geometry lives in the low-eigenvalue subspace: lowest-m modes preserve geodesics; a random-mode basis of equal size does not. The random curve is the mean over 25 draws, the band its full min–max envelope—even the best-case random draw stays far below the low-mode curve.

carries the ranks. Degree correction, by contrast, is *invisible* to the angle: dividing or multiplying each row by $\sqrt{k_i}$ (the diffusion-map convention and its opposite) leaves angle-ρ unchanged to three digits (0.889 either way), because a per-node radial factor cancels under row-normalization—a direct confirmation that the degree lives in the radius (Corollary 1), not the angle.

6 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is degree/density nuisance—von Luxburg-degenerate for the full, unnormalized commute radius at $d \geq 3$ (Corollary 1), and empirically degree-tracking for the truncated, normalized radius actually in use (§5.2). This gives Ng–Jordan–Weiss row-normalization a geodesic justification complementary to the cluster-separation theory of [23], and places it in the same scale-from-direction decomposition that recurs in vector and KV-cache compression [14] —the quotient, not the discard, being what recurs. The empirical reach of the basis across substrate families, its role as a manifold diagnostic, the compression scope boundary (angular quantization fails for attention keys, where scale is signal, not nuisance—condition (A2) of Theorem 7), and a scoped physical interpretation are developed in the companion Paper I.b [1].

Open problems. Proving Conjecture 3 (rank form, and the bi-Lipschitz strengthening) is now reduced, at fixed m, to the chart hypotheses (H1)–(H3) plus a distance-ratio condition for rank (Corollary 10, Proposition 13); the open core is the *scale-dependent* commute uniformity—fixed-scale uniformity in m is proven for the heat filter and refuted for the commute filter (Theorem 14, Proposition 15)—and graph transfer at growing spectral windows (Appendix C). Beyond the flat torus, where Corollary 11 settles them, (H1)–(H3) must be checked directly—the obstruction being the normalized upper bound, not the differential lower bound, which never collapses (Proposition 4). Also open: extending Corollary 1 to the truncated, normalized radius used in practice; synthesizing clean manifolds above $d \approx 2$; a dimension-adaptive basis that tunes mode-count and diffusion time against the fidelity decay; and whether the diagnostic transfers to learned representations (a preliminary probe is in the repository).

Reproducibility. The confirming experiments of this paper—the torus core result and controls, the theorem verification, the weighting ablation, and the truncation/normalization checks—are fully reproducible locally (NumPy/SciPy/NetworkX, CPU), each script emitting the exact committed result JSON in the accompanying repository, `https://github.com/ahb-sjsu/the-angular-observer` (pre-registered as `PREREG_RUNG3/4` with dated amendments). The extended empirical program and its reproducibility tiers are documented in the companion Paper I.b [1].

A Proof of Corollary 1

The diagonal statement $L^+_{ii} \to 1/k_i$ does not follow from the pairwise resistance limits alone: $R(i,j) = L^+_{ii} + L^+_{jj} - 2L^+_{ij}$ fixes only those combinations, so a summation argument must control the error it accumulates. Two inputs are used, both explicit in the corollary's statement: **(i)** the *uniform multiplicative* form of the von Luxburg–Radl–Hein degeneracy—for $d \geq 3$ graphs in their model there is a sequence $\eta_n \to 0$ with

$$R(i,j) = \left(\frac{1}{k_i} + \frac{1}{k_j}\right)(1 + \eta_{ij}), \qquad \sup_{i \neq j} |\eta_{ij}| \leq \eta_n$$

(this multiplicative-uniform form is not stated verbatim by [26] (their Corollaries 8–9 and Propositions 29–30): they prove an *additive* deviation bound on the rescaled commute distance, uniform over $i \neq j$, together with two-sided degree bounds; dividing the additive bound by $1/k_i + 1/k_j$ and using the degree bounds yields the displayed η_n); and **(ii)** a bounded degree ratio, $k_{\max}/k_{\min} \leq K$ for all n (their graph models satisfy this with high probability). Write $S := \sum_j 1/k_j$ and note $S/n \leq 1/k_{\min} \leq K/k_i$ for every i by (ii).

Step 1 (row sums). L^+ annihilates constants ($L\mathbf{1} = 0$ and L^+ shares the range of L on $\mathbf{1}^\perp$), so $\sum_j L^+_{ij} = 0$: the gauge. Summing $R(i,j)$ over $j \neq i$ and using the gauge,

$$\sum_{j \neq i} R(i,j) = (n-1)L^+_{ii} + \left(\operatorname{tr} L^+ - L^+_{ii}\right) - 2\sum_{j \neq i} L^+_{ij} = n\, L^+_{ii} + \operatorname{tr} L^+,$$

since $\sum_{j \neq i} L^+_{ij} = -L^+_{ii}$. By (i) the same sum equals

$$\sum_{j \neq i} \left(\frac{1}{k_i} + \frac{1}{k_j}\right)(1 + \eta_{ij}) = \frac{n-2}{k_i} + S + e_i, \qquad |e_i| \leq \eta_n \left(\frac{n}{k_i} + S\right).$$

Equating and dividing by n,

$$L^+_{ii} = \frac{1}{k_i} + C_n - \frac{2}{nk_i} + \frac{e_i}{n}, \qquad C_n := \frac{S - \operatorname{tr} L^+}{n} \quad \text{(independent of } i\text{)}. \tag{A.1}$$

Step 2 (trace identity kills C_n). Summing (A.1) over i: $\operatorname{tr} L^+ = S + nC_n - \frac{2}{n}S + \frac{1}{n}\sum_i e_i$. Substituting $nC_n = S - \operatorname{tr} L^+$ and solving,

$$\operatorname{tr} L^+ = S\left(1 - \frac{1}{n}\right) + \frac{1}{2n}\sum_i e_i, \qquad \text{hence} \qquad C_n = \frac{S}{n^2} - \frac{1}{2n^2}\sum_i e_i.$$

Since $|\sum_i e_i| \le \eta_n \sum_i (n/k_i + S) = 2\eta_n n S$, we get $|C_n| \le S/n^2 + \eta_n S/n \le (1/n + \eta_n)\, K/k_i$ for every i, using (ii).

Step 3 (conclusion). Feeding the bounds on C_n and e_i/n back into (A.1),

$$\Big|L^+_{ii} - \frac{1}{k_i}\Big| \;\le\; \frac{(1/n + \eta_n)K}{k_i} + \frac{2}{nk_i} + \frac{\eta_n}{k_i}\big(1 + K\big) \;\le\; \frac{C(K)\,\big(\eta_n + 1/n\big)}{k_i},$$

so $k_i L^+_{ii} \to 1$ uniformly over nodes, i.e. $L^+_{ii} = (1/k_i)(1+o(1))$ and $\|\Phi_i\| = \sqrt{L^+_{ii}} = (1/\sqrt{k_i})(1+o(1))$. $\square$

Without (ii) the conclusion weakens gracefully: the same computation gives $L^+_{ii} = 1/k_i + O\big((\eta_n + 1/n)\, S/n\big)$, an *additive* error at the harmonic-mean-degree scale, so the degeneracy survives at the pairwise level ($R \approx 1/k_i + 1/k_j$, which is (i)) even when per-node relative control is lost. The $j{=}i$ term ($R(i,i) = 0$ while $1/k_i + 1/k_j = 2/k_i$) is excluded from the sums, which is why the $(n-2)/k_i$ bookkeeping appears.

B Proof of Theorem 9

Setting and notation

M is a compact d-dimensional Riemannian manifold without boundary, volume normalized to 1; $(\mu_k, \varphi_k)_{k\ge 0}$ are the eigenpairs of its (positive) Laplace–Beltrami operator, $0 = \mu_0 < \mu_1 \le \mu_2 \le \cdots$, with $\{\varphi_k\}$ orthonormal in $L^2(M)$. Under non-uniform sampling with an admissible density the identical argument runs against the density-weighted operator that the graph Laplacian actually converges to [11, 6] (the $\alpha{=}1$ normalization of Paper I.b restores the geometric operator), with (H1)–(H2) read for its eigenpairs. The continuum objects are

$$F(x) = \Big(\tfrac{\varphi_k(x)}{\sqrt{\mu_k}}\Big)_{k=1}^{m}, \qquad R(x) = \|F(x)\|, \qquad N(x) = F(x)/R(x),$$

and m sits at a spectral gap: $\mu_m < \mu_{m+1}$. Write $\Phi_m := \max_{k\le m} \|\varphi_k\|_\infty < \infty$ (eigenfunctions on a smooth compact manifold are smooth). The sample is $x_1, \dots, x_n$ i.i.d. from the stated density class, G_n the ε-graph (or k-NN graph) on it, and (λ_k, v_k) the eigenpairs of L_{sym} with the empirical normalization $\frac{1}{n}\sum_i v_k(i)^2 = 1$; the algorithm forms $\Psi_i = (v_k(i)/\sqrt{\lambda_k})_{k=1}^m$ and $\hat u_i = \Psi_i/\|\Psi_i\|$. Alongside these, the *random-walk* eigenvectors $v_k^{\mathrm{rw}} := D^{-1/2} v_k$ are the eigenpairs of $L_{\mathrm{rw}} = D^{-1}(D - A) = D^{-1/2} L_{\mathrm{sym}} D^{1/2}$ at the *same* eigenvalue λ_k; we take them with the coordinated normalization $\frac{1}{n}\sum_i k_i\, v_k^{\mathrm{rw}}(i)^2 = 1$ (the empirical degree-weighted inner product—which is *identically* the algorithm's own L_{sym} constraint $\frac{1}{n}\sum_i v_k(i)^2 = 1$ viewed through $D^{1/2}$), under which each v_k^{rw} pairs mode-for-mode with the continuum eigenfunction φ_k orthonormal in the effective sampling measure (exactly $L^2(\rho)$ for uniform sampling or after the $\alpha{=}1$ reweighting of Paper I.b that restores the geometric operator; the ρ-versus-ρ^2 distinction under raw non-uniform sampling is immaterial, v_k^{rw} being only an intermediary to the identical angular map)—the convention in which the ∞-norm eigenvector rates below hold. Write $\Psi_i^{\mathrm{rw}} = (v_k^{\mathrm{rw}}(i)/\sqrt{\lambda_k})_{k=1}^m$.

Step 0: three invariances

The claim concerns only the pairwise distances $\|\hat u_i - \hat u_j\|$. These are invariant under (i) a global rescaling of all eigenvectors (the choice of normalization), (ii) a global rescaling of all eigenvalues (which rescales every weight $1/\sqrt{\lambda_k}$ by the same factor), and (iii) a fixed orthogonal transformation $Q \in O(m)$ applied to every Ψ_i (then $\|Q\hat u_i - Q\hat u_j\| = \|\hat u_i - \hat u_j\|$). We use (i) to work in the empirical normalization, (ii) to apply the deterministic graph-to-continuum eigenvalue scaling c_n of [6], and (iii) to absorb the basis alignment of Lemma 20.

(iv) Exact per-node degree cancellation. Since $v_k = D^{1/2} v_k^{\mathrm{rw}}$ applies the *same* node factor $\sqrt{k_i}$ in every mode k, the two commute-weighted embeddings differ only by a positive per-node scalar,

$$\Psi_i = \big(v_k(i)/\sqrt{\lambda_k}\big)_{k=1}^{m} = \sqrt{k_i}\,\big(v_k^{\mathrm{rw}}(i)/\sqrt{\lambda_k}\big)_{k=1}^{m} = \sqrt{k_i}\;\Psi_i^{\mathrm{rw}},$$

so row normalization returns the *identical* angular map $\hat{u}_i = \Psi_i^{\mathrm{rw}}/\|\Psi_i^{\mathrm{rw}}\|$—exactly, with no degree-ratio hypothesis. We may therefore calibrate the random-walk embedding Ψ_i^{rw}, whose ∞-norm convergence to the continuum is exactly the cited theorem, and read off the angle the algorithm computes from L_{sym}. This is what lets the theorem hold under non-uniform sampling: the node-dependent degree factor a symmetric calibration would have to control is cancelled by the normalization, not bounded.

Step 1: the continuum angular map is locally bi-Lipschitz

Lemma 18. *Under* (H1)–(H2)*, for all $x, y \in M$ with $d_M(x,y) \le r_0$,*

$$a\, d_M(x,y) \;\le\; \|N(x) - N(y)\| \;\le\; b\, d_M(x,y), \qquad a := \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{\rho^+}, \quad b := \frac{L_0}{\rho^-}.$$

Proof. Apply the identity of Lemma 5 to $u = F(x)$, $v = F(y)$. By (H1), $L_0^{-2} d_M^2 \le \|F(x) - F(y)\|^2 \le L_0^2\, d_M^2$; by (H2), $(R(x) - R(y))^2 \le \Lambda_0^2\, d_M^2$ and $(\rho^-)^2 \le R(x)R(y) \le (\rho^+)^2$. The ratio in the identity is therefore at least $(L_0^{-2} - \Lambda_0^2)\, d_M^2/(\rho^+)^2$—positive precisely because $\Lambda_0 < 1/L_0$—and at most $L_0^2\, d_M^2/(\rho^-)^2$. ☐

Note $a \le b$: a two-sided constant satisfies $L_0^{-1} \le L_0$, so $L_0 \ge 1$, hence $\sqrt{L_0^{-2} - \Lambda_0^2} \le L_0^{-1} \le L_0$, and $\rho^- \le \rho^+$.

Step 2: normalization is stable

Lemma 19. *For nonzero $u, w \in \mathbb{R}^m$,*

$$\left\|\frac{u}{\|u\|} - \frac{w}{\|w\|}\right\| \;\le\; \frac{2\,\|u - w\|}{\max(\|u\|, \|w\|)}.$$

Proof. $\frac{u}{\|u\|} - \frac{w}{\|w\|} = \frac{u-w}{\|u\|} + w\Big(\frac{1}{\|u\|} - \frac{1}{\|w\|}\Big)$, and the second term has norm $\big|\|w\| - \|u\|\big|/\|u\| \le \|u - w\|/\|u\|$. So the left side is at most $2\|u - w\|/\|u\|$; by symmetry in $u \leftrightarrow w$, also at most $2\|u - w\|/\|w\|$. ☐

Step 3: sup-norm spectral transfer

This is the step for which L^2 consistency [11] does not suffice and the ∞-norm rates of [6] are the right tool. Group $\mu_1 \le \cdots \le \mu_m$ into blocks of equal eigenvalues (multiplets); because m sits at a spectral gap, the last block ends exactly at m.

Throughout this step the calibration is carried out on the *random-walk* eigenvectors: to keep notation light we drop the superscript and write $v_k, \Psi_i, \widetilde{\Psi}_i$ for v_k^{rw} and the corresponding random-walk commute embeddings. By invariance (iv) the resulting angle is exactly the algorithm's $\hat{u}_i$, and the ∞-norm rates of [6]—stated for the unnormalized/random-walk eigenvectors—apply to v_k directly, with no degree-ratio hypothesis.

Lemma 20 (Calibration; external sup-norm transfer)**.** *Under the sampling model of [6, 7] there are sequences $\epsilon_n, \delta_n \to 0$—the eigenvector L^∞ rate of [7] (which establishes high-probability L^∞ and $C^{0,1}$ convergence of graph Laplacian eigenvectors to the Laplace–Beltrami eigenfunctions, with constants depending on the eigenvalue μ_m and hence on m; ϵ_n carries a factor $\sqrt{m}$), and the eigenvalue and gap-stability rate δ_n of [6]—and events $\mathcal{A}_n$ with $\mathbb{P}(\mathcal{A}_n) \to 1$ on which, after the eigenvalue scaling $\tilde{\lambda}_k := \lambda_k/c_n$ of invariance (ii), there is an orthogonal $Q_n \in O(m)$, block-diagonal across the multiplets, with*

$$\max_{k\le m} \left|\tilde{\lambda}_k - \mu_k\right| \le \delta_n, \qquad \max_{i\le n} \left\|Q_n v(i) - \varphi(x_i)\right\| \le \epsilon_n,$$

where $v(i) = (v_1(i), \ldots, v_m(i))$ and $\varphi(x) = (\varphi_1(x), \ldots, \varphi_m(x))$. Consequently, with $\widetilde{\Psi}_i := \big(v_k(i)/\sqrt{\tilde{\lambda}_k}\big)_{k=1}^m$ (the same angle as Ψ_i, by invariances (i),(ii),(iv)) and provided $\delta_n \le \mu_1/2$,

$$\max_{i\le n} \ \left\|Q_n \widetilde{\Psi}_i - F(x_i)\right\| \ \le \ E_n := \frac{\epsilon_n}{\sqrt{\mu_1}} + \frac{2\delta_n}{\mu_1^{3/2}} \sqrt{m} \left(\Phi_m + \epsilon_n\right) \ \longrightarrow \ 0.$$

In particular the graph radius inherits the continuum band: $\rho^- - E_n \le \|\widetilde{\Psi}_i\| \le \rho^+ + E_n$ for every node i.

Proof. (a) Spectral consistency. For the unnormalized and random-walk graph Laplacians, [6] give, with probability tending to one, eigenvalue convergence $|\tilde{\lambda}_k - \mu_k| \le \delta_n$ and the gap stability that makes "fixed m at a spectral gap" well-posed, while [7] supplies the eigenvector consistency in ∞-norm with explicit rates (and an approximate $C^{0,1}$/gradient rate $\epsilon_n' \to 0$ for the eigenvectors themselves, used for the cutoff in the remark below), stated per eigenvalue cluster. We make the block-orthogonal alignment explicit, since [7] (Theorem 2.6 with Remark 2.8) attaches to each discrete eigenvector v_k a *single* normalized limit eigenfunction $\tilde{\varphi}_k$ (their sup-norm rate, $\|v_k - \tilde{\varphi}_k\|_\infty \lesssim \mu_k \varepsilon_n$), not an orthonormal eigenspace basis. Within a multiplet the partners $\{\tilde{\varphi}_k\}$ need not be exactly orthonormal, but their *population* Gram matrix $G = \big(\langle \tilde{\varphi}_k, \tilde{\varphi}_l \rangle_{L^2(\rho)}\big)$ is $I + O(\varepsilon_n)$, routed through the discrete vectors: the v_k are orthonormal in the empirical inner product ($\frac{1}{n}\sum_i v_k(i)v_l(i) = \delta_{kl}$), each $\tilde{\varphi}_k$ is sup-norm ε_n-close to v_k on the sample, and empirical inner products concentrate on their $L^2(\rho)$ values at the eigenvector-consistency rate, so $\langle \tilde{\varphi}_k, \tilde{\varphi}_l \rangle_{L^2(\rho)} = \delta_{kl} + O(\varepsilon_n)$. Symmetric orthogonalization $G^{-1/2} = I + O(\varepsilon_n)$ therefore turns $\{\tilde{\varphi}_k\}$ into an exactly $L^2(\rho)$-orthonormal basis of the limiting eigenspace at cost $O(\varepsilon_n)$ in sup-norm; the alignment of this basis to the reference eigenbasis $\{\varphi_k\}$ is the *unique* block rotation $Q_n \in O(m)$ (block-diagonal across multiplets, the orthogonal factor in the polar decomposition of the cross-Gram $\langle \tilde{\varphi}_k, \varphi_l \rangle$)—not $G^{-1/2}$ itself, which is symmetric positive-definite. Each aligned discrete eigenvector is then $(\epsilon_n/\sqrt{m})$-close in sup-norm to its partner. (Multiplicity is exactly why the alignment is block-orthogonal rather than a per-mode sign choice; a simple eigenvalue is the 1×1 block, where it *is* a sign choice.) Collecting the m per-mode bounds into the vector $v(i)$ gives $\|Q_n v(i) - \varphi(x_i)\| \le \epsilon_n$.

(b) No normalization conversion is needed. The calibration in (a) is for the random-walk eigenvectors, and by the exact per-node cancellation of Step 0 (iv) the graph angular map is $\hat{u}_i = \Psi_i^{\mathrm{rw}}/\|\Psi_i^{\mathrm{rw}}\|$ regardless of whether L_{sym} or L_{rw} is diagonalized: the algorithm's embedding satisfies $\Psi_i = \sqrt{k_i}\,\Psi_i^{\mathrm{rw}}$ and the positive scalar $\sqrt{k_i}$ cancels under row normalization. In particular no uniform degree concentration is invoked and no bound on the degree ratio $k_i/\bar{k}_n$ is required, so the transfer is valid under non-uniform sampling; the sampling density enters only by fixing the limiting (weighted) operator whose eigenfunctions φ_k appear in F, as in the Setting. (The similarity $L_{\mathrm{rw}} = D^{-1/2} L_{\mathrm{sym}} D^{1/2}$ that underlies $v_k^{\mathrm{sym}} = D^{1/2} v_k^{\mathrm{rw}}$ is exact, so the shared eigenvalue λ_k used throughout is unaffected.)

(c) Bookkeeping. Let $W_\mu = \operatorname{diag}(\mu_k^{-1/2})$ and $\widetilde{W} = \operatorname{diag}(\tilde{\lambda}_k^{-1/2})$, so $F(x) = W_\mu \varphi(x)$ and $\widetilde{\Psi}_i = \widetilde{W} v(i)$. Since W_μ is a scalar multiple of the identity on each multiplet and Q_n is block-diagonal across the multiplets, $Q_n W_\mu = W_\mu Q_n$. Hence

$$Q_n \widetilde{\Psi}_i - F(x_i) = Q_n\big(\widetilde{W} - W_\mu\big)v(i) + W_\mu\big(Q_n v(i) - \varphi(x_i)\big).$$

For the first term, $\delta_n \le \mu_1/2$ gives $\tilde{\lambda}_k \ge \mu_1/2$, so

$$\big\|\widetilde{W} - W_\mu\big\|_{\mathrm{op}} = \max_{k \le m} \frac{|\mu_k - \tilde{\lambda}_k|}{\sqrt{\tilde{\lambda}_k\, \mu_k}\,\big(\sqrt{\tilde{\lambda}_k} + \sqrt{\mu_k}\big)} \;\le\; \frac{2\delta_n}{\mu_1^{3/2}},$$

while $\|v(i)\| \le \|Q_n v(i) - \varphi(x_i)\| + \|\varphi(x_i)\| \le \epsilon_n + \sqrt{m}\,\Phi_m \le \sqrt{m}\,(\Phi_m + \epsilon_n)$. For the second term, $\|W_\mu\|_{\mathrm{op}} = \mu_1^{-1/2}$. Adding the two bounds gives E_n. The radius statement is the reverse triangle inequality $\big|\|\widetilde{\Psi}_i\| - R(x_i)\big| = \big|\|Q_n\widetilde{\Psi}_i\| - \|F(x_i)\|\big| \le E_n$ together with $\rho^- \le R \le \rho^+$ from (H2). $\square$

General filters (proof of the calibration claim in Corollary 8). The proof above uses only two properties of the weight matrix $W_\mu = \operatorname{diag}(\mu_k^{-1/2})$: it is constant on multiplets (hence commutes with the block-orthogonal Q_n), and it is Lipschitz in the eigenvalue on the retained window. Both hold for any positive Lipschitz filter h: replacing W_μ by $H := \operatorname{diag}(h(\mu_k))$ and $\widetilde{W}$ by $\operatorname{diag}(h(\tilde{\lambda}_k))$, step (c) goes through verbatim with $\|\operatorname{diag}(h(\tilde{\lambda}_k)) - H\|_{\mathrm{op}} \le L_h \delta_n$ and $\|H\|_{\mathrm{op}} \le H_m$, giving $E_{n,h} = H_m \epsilon_n + L_h \delta_n \sqrt{m}\,(\Phi_m + \epsilon_n)$ as claimed; the commute filter $h(\lambda) = \lambda^{-1/2}$ (with $L_h \le 2\mu_1^{-3/2}$ on $[\mu_1/2, \infty)$ and $H_m = \mu_1^{-1/2}$) recovers the E_n displayed above.

Conditional structure, and the bridge to uniformity in m. Step 3 is *external*: it composes the named rate theorems [6, 7] into the calibration, and the transfer of the continuum bi-Lipschitz constants to the graph is then the elementary node-uniform perturbation of Step 4 (radius floor $\|\widetilde{\Psi}_i\| \ge \rho^- - E_n$; angular error $\le 2E_n/\rho^-$ by Lemma 19; cutoff $\theta_n \asymp E_n/a$). Consequently Theorem 9 is **unconditional in m for each fixed m**, resting only on this standard sup-norm spectral-transfer lemma—no self-contained spectral analysis is re-derived here. The one m-dependent seam is deliberate and worth stating plainly: the CGLewicka constants, hence ϵ_n and ϵ'_n, depend on the retained eigenfunctions' Lipschitz norms and on the spectral gap at μ_m, so the *rate* at which $\theta_n \to 0$ degrades as m grows. Making the transfer *uniform* in m—and with it (H1)—is exactly the frontier of Conjecture 3; we delimit it sharply in the discussion after Proposition 16, and it is filter-specific, since the $C^{0,1}$ rate ϵ'_n is what shrinks θ_n toward the connectivity scale (below which the twin obstruction lives).

Step 4: assembly

Proof of Theorem 9. Work on the event $\mathcal{A}_n$ of Lemma 20, with n large enough that $E_n < \rho^-/2$ (so $\widetilde{\Psi}_i \ne 0$ and every $\hat{u}_i$ is defined). By invariance (iii) we may compare angles through the calibrated embedding: writing $\hat{q}_i := Q_n\widetilde{\Psi}_i / \|Q_n\widetilde{\Psi}_i\| = Q_n \hat{u}_i$, Lemma 19 with $u = Q_n\widetilde{\Psi}_i$, $w = F(x_i)$, and $\|F(x_i)\| \ge \rho^-$ gives

$$\|\hat{q}_i - N(x_i)\| \;\le\; \frac{2E_n}{\rho^-} \qquad \text{for every node } i.$$

Set $\theta_n := 8E_n/(a\rho^-) \to 0$, and fix a sampled pair with $\theta_n \le d := d_M(x_i, x_j) \le r_0$. The pairwise angular error is at most $4E_n/\rho^- = (a/2)\,\theta_n \le (a/2)\,d$, so by Lemma 18, the triangle inequality, and $\|\hat{u}_i - \hat{u}_j\| = \|\hat{q}_i - \hat{q}_j\|$,

$$\|\hat{u}_i - \hat{u}_j\| \;\ge\; a\,d - \tfrac{a}{2}\,d \;=\; \tfrac{a}{2}\,d \;=\; A\,d, \qquad \|\hat{u}_i - \hat{u}_j\| \;\le\; b\,d + \tfrac{a}{2}\,d \;\le\; 2b\,d \;=\; B\,d,$$

the last inequality using $a \le b$ (Step 1). Both bounds hold simultaneously for *all* such pairs because the calibration of Lemma 20 is uniform over nodes, and $\mathbb{P}(\mathcal{A}_n) \to 1$. ☐

Step 5: from local to global (proof of Corollary 10)

Proof of Corollary 10. The map $(x, y) \mapsto \|N(x) - N(y)\|$ is continuous (F is continuous and $R \ge \rho^- > 0$) and, by (H3), strictly positive on the compact set $\{(x, y) \in M \times M : d_M(x, y) \ge r_0\}$; its minimum c_0 is therefore attained and positive. Work on the event of Lemma 20 with n large enough that $E_n < \rho^-/2$ and $4E_n/\rho^- \le c_0/2$. For sampled pairs with $\theta_n \le d_M \le r_0$, Theorem 9 gives the bounds with (A, B), hence with (A_g, B_g). For sampled pairs with $d := d_M > r_0$: the calibration bound $\|\hat{q}_i - N(x_i)\| \le 2E_n/\rho^-$ of Step 4 gives

$$\|\hat{u}_i - \hat{u}_j\| \;\ge\; \|N(x_i) - N(x_j)\| - \frac{4E_n}{\rho^-} \;\ge\; c_0 - \frac{c_0}{2} \;=\; \frac{c_0}{2} \;\ge\; \frac{c_0}{2\,\mathrm{diam}\, M}\, d \;\ge\; A_g\, d,$$

while unit vectors satisfy $\|\hat{u}_i - \hat{u}_j\| \le 2 \le (2/r_0)\, d \le B_g\, d$. ☐

Step 6: the flat torus (proof of Corollary 11)

Proof of Corollary 11. On $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$ with volume normalized to 1, the Laplace–Beltrami eigenvalues are $|k|^2$, $k \in \mathbb{Z}^2$: the first nontrivial eigenvalue $\mu = 1$ has multiplicity four, spanned by the L^2-orthonormal functions $\sqrt{2}\cos x$, $\sqrt{2}\sin x$, $\sqrt{2}\cos y$, $\sqrt{2}\sin y$, and the next eigenvalue is 2—so $m = 4$ exhausts the first multiplet and sits at a spectral gap. The truncated eigenmap (weights $1/\sqrt{\mu_k} = 1$) is the Clifford embedding $F(x, y) = \sqrt{2}\,(\cos x, \sin x, \cos y, \sin y)$.

(H2). $\|F\|^2 = 2(\cos^2 x + \sin^2 x) + 2(\cos^2 y + \sin^2 y) = 4$: the radius is constant, $R \equiv 2$, so $\Lambda_0 = 0 < 1/L_0$ and $\rho^- = \rho^+ = 2$.

(H1), globally. Using $(\cos a - \cos b)^2 + (\sin a - \sin b)^2 = 4\sin^2\!\big(\frac{a-b}{2}\big)$,

$$\|F(p) - F(q)\|^2 = 8\Big[\sin^2 \tfrac{\Delta x}{2} + \sin^2 \tfrac{\Delta y}{2}\Big], \qquad \Delta x, \Delta y \in [-\pi, \pi],$$

while $d_M(p, q)^2 = \Delta x^2 + \Delta y^2$. The elementary bounds $\frac{2}{\pi}|\delta| \le 2|\sin(\delta/2)| \le |\delta|$ on $[-\pi, \pi]$ give

$$\frac{2\sqrt{2}}{\pi}\, d_M(p, q) \;\le\; \|F(p) - F(q)\| \;\le\; \sqrt{2}\, d_M(p, q) \qquad \text{for } \textit{all } p, q,$$

so (H1) holds with $L_0 = \sqrt{2}$ (indeed $L_0^{-1} = 1/\sqrt{2} \le 2\sqrt{2}/\pi$) and $r_0 = \mathrm{diam}\, M = \pi\sqrt{2}$.

(H3). $N = F/2$; since the radius is constant, $N(p) = N(q)$ iff $F(p) = F(q)$ iff $e^{\mathrm{i}x_p} = e^{\mathrm{i}x_q}$ and $e^{\mathrm{i}y_p} = e^{\mathrm{i}y_q}$ iff $p = q$.

Because $r_0 = \mathrm{diam}\, M$, Theorem 9 alone is already global, with $A = \sqrt{L_0^{-2}/(2\rho^+)} = 1/(4\sqrt{2})$ and $B = 2L_0/\rho^- = \sqrt{2}$. The sharp continuum distortion of N is the exact ratio range above: $\|N(p) - N(q)\|/d_M(p, q) \in [\sqrt{2}/\pi,\ 1/\sqrt{2}]$, the ratio of whose extremes is $\pi/2$, attained along a coordinate direction at antipodal ($|\delta| = \pi$) and infinitesimal separations respectively. Basis-independence: any L^2-orthonormal basis of the $\mu = 1$ eigenspace equals Q applied to the Clifford one for some $Q \in O(4)$, and angular pairwise distances are Q-invariant (Step 0, invariance (iii)). ☐

Step 7: the small-ball ratio law (used by Proposition 13)

Let D_1, D_2 be i.i.d. with the small-ball distance law $F(r) = (r/r_0)^d$ on $[0, r_0]$ (density $\propto r^{d-1}$, the flat approximation to pair distances within a geodesic ball). For $\kappa \ge 1$,

$$\mathbb{P}\big(D_2 \ge \kappa D_1\big) = \int_0^{r_0} F\big(\tfrac{y}{\kappa}\big)\, dF(y) = \frac{1}{\kappa^d} \int_0^{r_0} \Big(\frac{y}{r_0}\Big)^d \frac{d\, y^{d-1}}{r_0^d}\, dy = \frac{1}{2\kappa^d},$$

so by symmetry $\mathbb{P}(\max / \min \geq \kappa) = \kappa^{-d}$ and $\mu(\kappa) = 1 - \kappa^{-d}$ exactly. The rank certificate of Proposition 13 is thus $\tau \geq 2\kappa^{-d} - 1$, $\rho_S \geq 3\kappa^{-d} - 2$: exponential decay of the worst case in the intrinsic dimension.

Why the resolution cutoff is necessary

The cutoff θ_n is forced by the statement's generality, not by the method. In a hard-threshold ε-graph, two sample points at distance below the bandwidth can be *combinatorial twins*—identical neighbourhoods, hence identical rows of A and equal degrees. For twins i, j the difference $e_i - e_j$ is itself an eigenvector of L_{sym}—at eigenvalue 1 for non-adjacent twins and $1+1/k$ for adjacent ones—so every eigenvector at any *other* eigenvalue is orthogonal to it and has identical entries at i and j. In particular every low mode does, hence $\hat{u}_i = \hat{u}_j$ exactly while $d_M(x_i, x_j) > 0$: no lower bound $A\, d_M$ can hold below the connectivity scale. ("Identical in every eigenvector" would be literally false—the localized twin-difference mode itself separates them—but that mode sits at the top of the spectrum, far above anything the angular map retains.) What is improvable is the *size* of θ_n: our proof takes it at the sup-norm consistency rate E_n, much larger than the bandwidth; a C^1 (gradient) consistency argument—the Lipschitz regularity theory for graph-Laplacian eigenfunctions of [7] is the natural ingredient—would control eigenvector *differences* at small separations directly and could shrink θ_n toward the connectivity scale, below which the twin obstruction is final.

Two further remarks. (1) Every constant in E_n depends on the fixed mode-count m (through $\sqrt{m}$, Φ_m, μ_1, and the gap $\mu_{m+1} - \mu_m$); nothing here is claimed uniform in m. The uniformity in m observed empirically (§5.2) corresponds to the uniformity of (H1) [2, 19] and is exactly what separates Theorem 9 from the full Conjecture 3; Appendix C settles its continuum side by filter (uniform for heat, scale-dependent for commute). (2) The theorem is local ($d_M \leq r_0$) because (H1) is; the global statement needs precisely the injectivity hypothesis (H3), which Corollary 10 (Step 5) carries out—and on the flat torus (H1) is itself global, so Corollary 11 (Step 6) needs none of (H1)–(H3).

C Uniformity in the truncation

This appendix proves Theorem 14 and Proposition 15, and records the growing-window graph-transfer conditions. Throughout, volume is normalized to 1, p_t is the heat kernel of M, and cutoffs are multiplet-closed.

Proof of Theorem 14

Let $F_t(x) = \big(e^{-t\mu_k/2}\varphi_k(x)\big)_{k\geq 1} \in \ell^2$ be the *full* heat embedding with the constant mode removed; its kernel is

$$K_t(x, y) := \langle F_t(x), F_t(y)\rangle = p_t(x, y) - 1.$$

Step 1: N_t is a uniformly noncollapsing immersion for small t. For a unit tangent vector $v \in T_x M$,

$$\|dF_t(v)\|^2 = \partial_x^v \partial_y^v K_t(x, y)\big|_{y=x}, \qquad dR_t(v) = \frac{\partial_v K_t(x, x)}{2R_t(x)}, \qquad R_t^2(x) = p_t(x, x) - 1.$$

By the Minakshisundaram–Pleijel expansion (cf. [5]), $p_t(x, x) = (4\pi t)^{-d/2}\big(1 + t\,\mathrm{scal}(x)/6 + O(t^2)\big)$ and $\partial_x^v \partial_y^v p_t\big|_{y=x} = \frac{1}{2t}(4\pi t)^{-d/2}(1 + O(t))$, both uniformly on compact M. The only x-dependence of the diagonal at this order is through the curvature term, so $\partial_v p_t(x, x) = (4\pi t)^{-d/2}\, O(t)$, whence

$dR_t(v)^2 = (4\pi t)^{-d/2}\, O(t^2)\cdot O(1)$, which is o of the derivative energy $\asymp t^{-1}(4\pi t)^{-d/2}$. Therefore there is $t_0(M) > 0$ such that for $t \le t_0$ the tangential square $T_t(x,v)^2 := \|dF_t(v)\|^2 - dR_t(v)^2$ obeys

$$T_t(x,v)^2 \;\ge\; c\,t^{-d/2-1}, \qquad R_t(x)^2 \asymp t^{-d/2}, \qquad \text{hence} \qquad \|dN_t(v)\|^2 = \frac{T_t(x,v)^2}{R_t(x)^2} \;\ge\; \frac{c'}{t},$$

uniformly over the unit tangent bundle (the last identity is Proposition 6).

Step 2: the truncation tail vanishes in C^2. Eigenfunction Hölder/derivative norms grow only polynomially, $\|\varphi_k\|_{C^r} \le C\mu_k^{(d-1)/4+r/2}$ [30], while the heat weights decay exponentially; hence $\|F_t - F_{t,m}\|_{C^2} \to 0$ as $m \to \infty$ (tail sums $\sum_{k>m} e^{-t\mu_k}\mu_k^{c(d,r)} \to 0$ by Weyl's law). Consequently $T_{t,m} \to T_t$, $R_{t,m} \to R_t$, and the C^2-norms of $N_{t,m}$ converge, all uniformly; choose $m_0(t)$ so that for $m \ge m_0(t)$, $T_{t,m}^2 \ge \frac{1}{2}c\,t^{-d/2-1}$, $R_{t,m} \asymp t^{-d/4}$, and $\sup_{m\ge m_0}\|N_{t,m}\|_{C^2} =: C_2(t) < \infty$.

Step 3: from uniform immersion to uniform bi-Lipschitz. Let $a'_t := \inf_{m\ge m_0}\inf_{x,|v|=1}\|dN_{t,m}(v)\| > 0$ from Steps 1–2. For x,y with $d := d_M(x,y) \le r_t := a'_t/(2C_2(t))$, Taylor along a minimizing geodesic γ gives

$$\|N_{t,m}(x) - N_{t,m}(y)\| \;\ge\; \|dN_{t,m}(\gamma'(0))\|\,d - \tfrac{1}{2}C_2(t)\,d^2 \;\ge\; \Big(a'_t - \tfrac{1}{2}C_2(t)\,r_t\Big)d \;\ge\; \tfrac{3a'_t}{4}\,d,$$

and the mean-value bound gives $\|N_{t,m}(x) - N_{t,m}(y)\| \le b_t\,d$ with $b_t := \sup_{m\ge m_0}\mathrm{Lip}(N_{t,m}) < \infty$. All constants depend on t and M only—not on m. $\square$

Lemma 21 (Differentiated pointwise Weyl law)**.** *Let (M,g) be a smooth closed Riemannian d-manifold with orthonormal Laplace eigenfunctions φ_k, eigenvalues μ_k. There are constants $c_d, c'_d > 0$ depending only on (M,g) such that, uniformly in $x \in M$,*

$$\sum_{0<\mu_k\le\Lambda}\varphi_k(x)^2 \;=\; c_d\,\Lambda^{d/2}\big(1+O(\Lambda^{-1/2})\big), \qquad \sum_{0<\mu_k\le\Lambda}\|\nabla\varphi_k(x)\|^2 \;=\; c'_d\,\Lambda^{(d+2)/2}\big(1+O(\Lambda^{-1/2})\big),$$

with spatially constant leading coefficients.

Proof. Both are on-diagonal statements about the spectral projector $\Pi_\lambda(x,y) = \sum_{\sqrt{\mu_k}\le\lambda}\varphi_k(x)\varphi_k(y)$ at frequency $\lambda = \sqrt{\Lambda}$: the first is Hörmander's pointwise Weyl law with sharp remainder [30], $\Pi_\lambda(x,x) = \tilde{c}_d\lambda^d + O(\lambda^{d-1})$; the second is the mixed-derivative case $\partial_x\cdot\partial_y\Pi_\lambda(x,y)\big|_{y=x} = \tilde{c}'_d\lambda^{d+2} + O(\lambda^{d+1})$, the uniform derivative estimates for the spectral function on a closed manifold of [28] (one power of λ below the main term, uniformly on M; no non-self-focal restriction, which the finer scaling limit of [29] would carry). Substituting $\lambda = \sqrt{\Lambda}$ gives the stated relative remainders. $\square$

Proof of Proposition 15

Let $E_\Lambda(x,y) = \sum_{\mu_k\le\Lambda}\varphi_k(x)\varphi_k(y)$ be the spectral function. The pointwise Weyl law with uniform remainder [30] gives $E_\Lambda(x,x) = C_d\Lambda^{d/2} + O(\Lambda^{(d-1)/2})$ with spatially constant leading coefficient, and its differentiated form (the uniform derivative-Weyl estimates of [28]; equivalently Lemma 21) gives, uniformly in x and unit v,

$$\partial_x^v\partial_y^v E_\Lambda(x,y)\big|_{y=x} = C'_d\,\Lambda^{d/2+1}\big(1+o(1)\big).$$

Abel summation in the cutoff converts these into the commute-weighted sums:

$$\|dF_\Lambda(v)\|^2 = \sum_{0<\mu_k\le\Lambda}\mu_k^{-1}\big(\partial_v\varphi_k(x)\big)^2 = c_d\,\Lambda^{d/2}\big(1+o(1)\big),$$

$$G_\Lambda(x,x) := R_\Lambda^2(x) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)^2 = \begin{cases} C_d\,\Lambda^{d/2-1}\big(1+o(1)\big), & d>2,\\ C_2\log\Lambda\,\big(1+o(1)\big), & d=2,\end{cases}$$

again uniformly in x (Lemma 21 in the commute-weighted normalization; Abel summation in the cutoff converts the on-diagonal projector asymptotics into the μ^{-1}-weighted sums), with spatially constant leading coefficients. Because the leading coefficient of $G_\Lambda(x,x)$ is spatially constant, its spatial derivative has no leading term, and the differentiated remainder estimates of [28] bound what remains, $\partial_v G_\Lambda(x,x) = o(\Lambda^{(d-1)/2})$ uniformly—weaker than the naive $o(\Lambda^{d/2-1})$ and all that is needed—so the radial term $dR_\Lambda(v)^2 = \big(\partial_v G_\Lambda(x,x)\big)^2/\big(4G_\Lambda(x,x)\big) = o(\Lambda^{d/2})$ is lower order than the derivative energy. Hence $T_\Lambda(x,v)^2 \ge c\,\Lambda^{d/2}$ for $\Lambda\ge\Lambda_0$, uniformly, which is the display; dividing by R_Λ^2 gives $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$ ($d>2$) and $\sqrt{\Lambda/\log\Lambda}$ ($d=2$).

The S^1 computation, and the failure of fixed-scale uniformity. On S^1 ($\mu=k^2$, whole eigenspaces, weights $\mu^{-1/2}=1/k$),

$$F_N(x) = \Big(\tfrac{\cos kx}{k},\tfrac{\sin kx}{k}\Big)_{k=1}^N: \qquad R_N^2 = \sum_{k\le N}\frac{1}{k^2}\ \text{(constant, bounded)}, \qquad \|dF_N\|^2 = \sum_{k\le N} 1 = N,$$

so $\|dN_N\| = \|dF_N\|/R_N \asymp \sqrt{N}$ (consistent with the general formula: $\Lambda=N^2$, $d=1$, $T_\Lambda/R\asymp\Lambda^{1/4}$). A pair at distance $\sim 1/N$ is stretched by $\sim\sqrt{N}$, so any upper Lipschitz constant valid on a *fixed* ball $\{d\le r_0\}$ must grow like $\sqrt{m}$: no m-independent fixed-scale bi-Lipschitz theorem can hold for the commute filter, on any manifold (the same growth holds in all d). The lower bound at fixed scale, by contrast, survives on S^1 (project onto the first multiplet, whose weight is m-independent while R_N is bounded)—the failure is one-sided and lives below the wavelength $\Lambda^{-1/2}$. □

Remark (Cutoff-insensitivity). The conclusion of Proposition 15 does not depend on the sharp spectral cutoff: replacing $\sum_{\mu_k\le\Lambda}$ by a smoothed Littlewood–Paley weight $\chi(\mu_k/\Lambda)$ yields the same growth orders with full asymptotic expansions (smoothed spectral sums admit wave-kernel expansions without Tauberian remainders), and on $\mathbb{S}^1$ the sharp-cutoff computation is exact with no Weyl input at all. The proposition's content—no m-independent fixed-scale upper Lipschitz constant for the commute filter—is cutoff-independent.

A uniform lower angular bound on flat tori

Proposition 22 (Uniform lower angular bound on flat tori)**.** *Let $T=\mathbb{R}^d/\Gamma$ be a fixed flat torus with dual lattice Γ^*. For a multiplet-closed spectral cutoff Λ, set*

$$F_\Lambda(x) = \Big(\tfrac{e^{i\langle\xi,x\rangle}}{|\xi|}\Big)_{\xi\in\Gamma^*\setminus\{0\},\ |\xi|^2\le\Lambda}, \qquad N_\Lambda(x) = \frac{F_\Lambda(x)}{\|F_\Lambda(x)\|}$$

(equivalently, pair ξ with $-\xi$ and use real sine–cosine coordinates). There exist $\Lambda_0(T)<\infty$ and $c_T>0$ such that

$$\|N_\Lambda(x)-N_\Lambda(y)\| \ \ge\ c_T\, d_T(x,y)$$

for every multiplet-closed $\Lambda\ge\Lambda_0(T)$ and every $x,y\in T$; the lower angular Lipschitz constant is thus uniform in the mode count $m=m(\Lambda)$. For the standard torus $\mathbb{T}^d=\mathbb{R}^d/(2\pi\mathbb{Z})^d$ one may take $\Lambda_0=1$.

Proof. We work first on the standard torus; the general lattice is treated at the end. Write $R=\sqrt{\Lambda}$ and $\mathcal{K}_R=\{k\in\mathbb{Z}^d\setminus\{0\} : |k|\le R\}$. Because the cutoff contains complete sine–cosine pairs,

$S_R := \|F_R(x)\|^2 = \sum_{k\in\mathcal{K}_R} |k|^{-2}$ is independent of x. If $z = x - y$ is represented in $[-\pi,\pi]^d$ then $d_{\mathbb{T}^d}(x,y) = |z|$ and

$$\|N_R(x) - N_R(y)\|^2 = \frac{2A_R(z)}{S_R}, \qquad A_R(z) := \sum_{k\in\mathcal{K}_R} \frac{1-\cos(k\cdot z)}{|k|^2}. \tag{1}$$

We prove $A_R(z) \geq c_d |z|^2 S_R$ with $c_d > 0$ independent of R.

A Fourier-block estimate. For $q \geq 2$ and $t \in [-\pi,\pi]$ set $a_q(t) := \frac{1}{q}\sum_{n=q}^{2q-1} \cos(nt)$. There is an absolute $c > 0$ with

$$1 - |a_q(t)| \;\geq\; c\,\min\{1,\, q^2t^2\}. \tag{2}$$

Indeed, if U is uniform on $\{q,\dots,2q-1\}$ and $\phi_q(t) = \mathbb{E}\, e^{iUt}$, then $|a_q(t)| \leq |\phi_q(t)|$. When $q|t| \leq 1$, for an independent copy U',

$$1 - |\phi_q(t)| \;\geq\; \tfrac12\big(1 - |\phi_q(t)|^2\big) = \tfrac12\,\mathbb{E}\big[1 - \cos((U-U')t)\big] \;\geq\; c\,q^2t^2,$$

using $1 - \cos s \geq cs^2$ for $|s| \leq 1$ and $\mathbb{E}(U-U')^2 \asymp q^2$. When $q|t| \geq 1$, the exact formula $|\phi_q(t)| = |\sin(qt/2)|/(q|\sin(t/2)|)$ is bounded by a universal $\rho < 1$: for bounded q the ratio is < 1 on the compact locus $q|t| \geq 1$, and as $q \to \infty$ it tends to 0 unless $t \to 0$, where with $s = qt/2 \geq \frac12$ it equals $|\sin s|/s < 1$. This gives (2).

Lattice blocks. Choose j with $|z_j| = \|z\|_\infty$. For dyadic q put $B_{q,j} = \{k \in \mathbb{Z}^d : q \leq |k_j| < 2q,\ |k_\ell| < q\ (\ell \neq j)\}$, disjoint across dyadic q, with $|B_{q,j}| \asymp_d q^d$ and $|k|^2 \leq (d+3)q^2$ on $B_{q,j}$. The block average of $\cos(k\cdot z)$ factors as $a_q(z_j)\prod_{\ell\neq j} b_q(z_\ell)$ with each normalized symmetric Dirichlet sum $|b_q| \leq 1$, so by (2)

$$\frac{1}{|B_{q,j}|}\sum_{k\in B_{q,j}} \big(1 - \cos(k\cdot z)\big) \;\geq\; 1 - |a_q(z_j)| \;\geq\; c\,\min\{1, q^2z_j^2\}.$$

With $|k|^2 \leq (d+3)q^2$, $|B_{q,j}| \asymp q^d$, and $\min\{1, q^2z_j^2\} \geq |z|^2/(d\pi^2)$ (from $z_j^2 \geq |z|^2/d$ and $|z| \leq \sqrt{d}\,\pi$),

$$\sum_{k\in B_{q,j}} \frac{1-\cos(k\cdot z)}{|k|^2} \;\geq\; c_d\, q^{d-2}|z|^2.$$

Every block with $q \leq R/\sqrt{d+3}$ lies in $\mathcal{K}_R$; summing over dyadic q in that range, $A_R(z) \geq c_d|z|^2 \sum_{q\ \text{dyadic},\, 2\leq q\leq R/\sqrt{d+3}} q^{d-2}$, while a dyadic decomposition of the lattice ball gives $S_R \leq C_d\big(1 + \sum_{q\ \text{dyadic},\, q\leq R} q^{d-2}\big)$. The two sums are comparable for large R—bounded for $d = 1$, $\asymp \log R$ for $d = 2$, $\asymp R^{d-2}$ for $d \geq 3$—so $A_R(z) \geq c_d|z|^2 S_R$. For the finitely many smaller cutoffs the first multiplet already gives $A_R(z) \geq \sum_j (1 - \cos z_j) \geq c_d|z|^2$ with S_R bounded, so the bound holds for all $R \geq 1$. Substituting into (1), $\|N_R(x) - N_R(y)\|^2 \geq c_d|z|^2 = c_d\, d_{\mathbb{T}^d}(x,y)^2$.

For a general flat torus, a dual-lattice basis $b_1,\dots,b_d$ and norm equivalence $c_T|n| \leq |\xi(n)| \leq C_T|n|$ reduce the estimate to the standard case in angular coordinates $\theta \in [-\pi,\pi]^d$, with constants depending on T; taking $\Lambda_0(T)$ large enough to contain a dual basis handles the initial cutoffs. $\square$

Together with Proposition 15 this locates the commute frontier exactly: $c_T\, d_T(x,y) \leq \|N_m(x) - N_m(y)\|$ uniformly in m, while there is *no* m-uniform upper bound at a fixed spatial scale, since $\|dN_m\|$ grows with the cutoff. The lower Lipschitz half of the metric conjecture is thus proved on the reference substrate family; only the scale-dependent upper half remains.

Remark (Exact upper-Lipschitz divergence). The companion upper bound diverges, on $\mathbb{T}^d$ exactly and for every d. Because S_R is independent of x, normalization produces no radial cancellation, so for a coordinate direction e_j,

$$\|dN_R(e_j)\|^2 = \frac{1}{S_R}\sum_{k\in\mathcal{K}_R}\frac{k_j^2}{|k|^2} = \frac{|\mathcal{K}_R|}{d\,S_R},$$

using $\sum_{j=1}^d k_j^2/|k|^2 = 1$ at every lattice point. With $|\mathcal{K}_R| \asymp R^d$ and the orders of S_R above,

$$\mathrm{Lip}(N_R) \asymp \begin{cases} R^{1/2}, & d=1,\\ R/\sqrt{\log R}, & d=2,\\ R, & d\geq 3,\end{cases}$$

so no R-independent upper Lipschitz constant exists on any fixed geodesic ball. This is an exact obstruction, not a gap in technique, and it survives a global output rescaling $N_R \mapsto c_R N_R$: uniform upper control forces $c_R \to 0$, which collapses the fixed-diameter image (diam $\leq 2c_R \to 0$) and destroys the lower bound. The commute map is therefore *uniformly co-Lipschitz with a quantitatively diverging upper Lipschitz constant*; genuine two-sided uniformity requires a faster filter—the heat weighting (Theorem 14), or amplitude $|k|^{-s}$ with $s > (d+2)/2$, for which the numerator above converges while S_R tends to a positive constant.

A spectral-filter phase diagram

The commute weighting is one member of the fractional family

$$F_{\Lambda,s}(x) = \big(\mu_k^{-s/2}\varphi_k(x)\big)_{0<\mu_k\leq\Lambda}, \qquad K_{\Lambda,s}(x,y) = \sum_{0<\mu_k\leq\Lambda}\mu_k^{-s}\varphi_k(x)\varphi_k(y),$$

with commute at $s=1$, the plain eigenmap at $s=0$, and the heat weighting a rapidly decaying alternative. Weyl's law $\mu_k \asymp k^{2/d}$ separates three regimes by three convergence thresholds.

Proposition 23 (Filter thresholds)**.** *As $\Lambda\to\infty$,*

$$\sum_k \mu_k^{-2s} < \infty \iff s > \tfrac{d}{4}, \qquad \sum_k \mu_k^{-s} < \infty \iff s > \tfrac{d}{2}, \qquad \sum_k \mu_k^{1-s} < \infty \iff s > \tfrac{d}{2}+1,$$

controlling respectively: (i) the $L^2(M\times M)$ (Hilbert–Schmidt) limit of the correction kernel $K_{\Lambda,s}$—hence stability of the rank information it carries; (ii) the trace $\int_M \|F_{\Lambda,s}\|^2 = \sum_k \mu_k^{-s}$, i.e. finiteness of the radius in the infinite embedding; and (iii) the differential energy $\int_M \|dF_{\Lambda,s}\|^2 = \sum_k \mu_k^{1-s}$, i.e. a finite uniform upper Lipschitz constant.

Proof. Each equivalence is Weyl's law: $\mu_k \asymp k^{2/d}$ gives $\mu_k^{-a} \asymp k^{-2a/d}$, and $\sum_k k^{-2a/d} < \infty \iff 2a/d > 1$; take $a = 2s, s, s-1$. For (i), orthonormality gives $\|K_{\Lambda,s}\|^2_{L^2(M\times M)} = \sum_{\mu_k\leq\Lambda}\mu_k^{-2s}$, Cauchy in Λ iff the sum converges. Claims (ii)–(iii) integrate $\|F\|^2$ and $\|dF\|^2$ over M. □

For the commute filter $s=1$ the thresholds read $d<4$, $d<2$, and $d<0$:

Property	Convergent sum	Commute ($s=1$)
Rank kernel Hilbert–Schmidt stable	$\sum \mu_k^{-2}$	holds for $d\leq 3$
Radius finite in the limit	$\sum \mu_k^{-1}$	only $d=1$ (diverges $d\geq 2$)
Uniform upper Lipschitz bound	$\sum \mu_k^{0}$	fails $\forall\, d\geq 1$

This resolves three observations at once: rank stability can survive in $d = 1, 2, 3$ (the empirical "flat in m"); the radius changes character at $d = 2$ (Corollary 1); and two-sided uniform metric control fails in every positive dimension (Proposition 15, and the exact torus divergence of the remark above). The commute filter sits at the *critical* exponent for the rank kernel, whose stability boundary is the dimension $d = 4$—a sharper structural prediction than the five-family linear trend of Paper I.b. A filter with $s > d/2 + 1$ (the heat weighting, or amplitude $\mu_k^{-s/2}$ with $s > d/2 + 1$) restores a uniform two-sided bound.

A rank-limit theorem on flat tori ($d \leq 3$)

On a flat torus the diagonal kernel $S_\Lambda = \|F_\Lambda(x)\|^2$ is independent of x, so the angular distance is an exact decreasing function of a single scalar:

$$\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - \frac{2\,G_\Lambda(x-y)}{S_\Lambda}, \qquad G_\Lambda(z) = \sum_{0<|k|^2\leq\Lambda} \frac{\cos(k\cdot z)}{|k|^2}.$$

Because S_Λ does not depend on the pair, the angular distances carry exactly the reverse ranking of $G_\Lambda(x-y)$—so although every distance concentrates at $\sqrt{2}$ as $\Lambda \to \infty$ (the radius diverges, $S_\Lambda \to \infty$ for $d \geq 2$), the *ordering* that rank statistics read need not degenerate.

Proposition 24 (Rank stability on flat tori)**.** *Let $d \leq 3$. Then $\sum_{k\neq 0}|k|^{-4} < \infty$, so $G_\Lambda \to G$ in $L^2(\mathbb{T}^d)$, where G is the zero-mean periodic Green's function. If (X, Y) is drawn from an absolutely continuous pair law (uniform sampling suffices) whose induced law of $G(X-Y)$ is atomless, then the population Spearman and Kendall correlations between $d_{\mathbb{T}^d}(X, Y)$ and $\|N_\Lambda(X) - N_\Lambda(Y)\|$ converge to those of $\big(d_{\mathbb{T}^d}(X, Y),\, -G(X-Y)\big)$. On $\mathbb{S}^1$ the periodic Green's function is strictly monotone in geodesic separation, so the limiting rank correlation is exactly* 1.

Proof. G has Fourier coefficients $|k|^{-2}$, so $\|G_\Lambda - G\|_{L^2}^2 = \sum_{|k|^2>\Lambda}|k|^{-4} \to 0$ exactly when $\sum_k |k|^{-4} < \infty$, i.e. $4 > d$. Hence $G_\Lambda(X-Y) \to G(X-Y)$ in probability along the pair law. The angular distance is a fixed continuous strictly decreasing function of $G_\Lambda(X-Y)$ (the common factor $S_\Lambda > 0$ is immaterial to order), so its ranks converge to those of $-G(X-Y)$; atomlessness of the limit law makes the Spearman and Kendall functionals continuous there, giving convergence. On $\mathbb{S}^1$, $G(\theta) = \sum_{k\geq 1}\cos(k\theta)/k^2$ is strictly decreasing on $[0, \pi]$ (its derivative $-\sum_{k\geq 1}\sin(k\theta)/k = (\theta - \pi)/2 < 0$ there), so $-G$ is strictly increasing in the geodesic distance $|\theta|$ and the rank correlation is 1. ☐

This upgrades the empirical "flat in m" (§5.2) to a rank-stability theorem: for $d \leq 3$ the commute angular ranking converges to a fixed Green-kernel ranking—exact on $\mathbb{S}^1$, and a computable limiting quantity on $\mathbb{T}^2, \mathbb{T}^3$ (where G is direction-dependent, so the limit is below 1). For $d \geq 4$ the kernel is not L^2-stable (Proposition 23) and no such limit is claimed.

The general Green-kernel rank limit ($d \in \{2, 3\}$)

The flat-torus computation generalizes to every closed manifold of dimension 2 or 3, bypassing the impossible uniform upper Lipschitz bound and isolating the exact geometric obstruction. For independent X, Y from the volume measure, define the *Green-rank coefficient* $\rho_G(M) := \rho_S\big(d_M(X, Y), -G_M(X, Y)\big)$, where G_M is the zero-mean Green kernel of the Laplace–Beltrami operator. Angular Preservation then splits into two questions: does the truncated angular ranking converge uniformly in m to the Green-kernel ranking, and is $\rho_G(M) > 0$?

Theorem 25 (Green-kernel limit of the commute angular ranking). *Let M be a closed connected Riemannian manifold of dimension $d \in \{2,3\}$, volume normalized to 1, with Laplace–Beltrami eigenpairs (μ_k, φ_k). At a multiplet-closed cutoff Λ set $G_\Lambda(x,y) = \sum_{0<\mu_k\le\Lambda} \varphi_k(x)\varphi_k(y)/\mu_k$, $S_\Lambda(x) = G_\Lambda(x,x)$, and $N_\Lambda(x) = (\mu_k^{-1/2}\varphi_k(x))_{0<\mu_k\le\Lambda}/\sqrt{S_\Lambda(x)}$. Let (X,Y) have bounded density with respect to $\mathrm{vol}\otimes\mathrm{vol}$, with the laws of $d_M(X,Y)$ and $G(X,Y)$ atomless, where $G(x,y) = \sum_{k\ge1}\varphi_k(x)\varphi_k(y)/\mu_k \in L^2(M\times M)$. Then*

$$\rho_S\big(d_M(X,Y),\ \|N_\Lambda(X)-N_\Lambda(Y)\|\big) \ \longrightarrow\ \rho_S\big(d_M(X,Y),\ -G(X,Y)\big),$$

and likewise for Kendall's τ. Consequently, if $\rho_G(M) > 0$ there are $\Lambda_0 < \infty$ and $\rho_0 > 0$ with $\rho_S\big(d_M(X,Y), \|N_\Lambda(X)-N_\Lambda(Y)\|\big) \ge \rho_0$ for every multiplet-closed $\Lambda \ge \Lambda_0$: the rank form of Angular Preservation is uniform in the mode count beyond a finite cutoff.

Proof. Write $C_\Lambda(x,y) = \langle N_\Lambda(x), N_\Lambda(y)\rangle = G_\Lambda(x,y)/\sqrt{S_\Lambda(x)S_\Lambda(y)}$; since $\|N_\Lambda(x)-N_\Lambda(y)\|^2 = 2 - 2C_\Lambda(x,y)$, angular distance carries the reverse ranking of C_Λ. *(1) Hilbert–Schmidt limit.* The $\varphi_k\otimes\varphi_k$ are orthonormal in $L^2(M\times M)$, so $\|G_{\Lambda'}-G_\Lambda\|_{L^2}^2 = \sum_{\Lambda<\mu_k\le\Lambda'}\mu_k^{-2}$; by Weyl's law $\mu_k \asymp k^{2/d}$ this converges iff $d<4$, so $G_\Lambda \to G$ in $L^2(M\times M)$ for $d = 2,3$. *(2) The diagonal is asymptotically constant.* The uniform pointwise Weyl law $\sum_{\mu_k\le L}\varphi_k(x)^2 = c_d L^{d/2} + O(L^{(d-1)/2})$ and partial summation give, uniformly in x, $S_\Lambda(x) = a_2\log\Lambda + O(1)$ $(d=2)$ or $a_3\Lambda^{1/2} + O(\log\Lambda)$ $(d=3)$; writing A_Λ for the leading term, $\sup_x|S_\Lambda(x)/A_\Lambda - 1| \to 0$. *(3) Rescale.* $H_\Lambda := A_\Lambda C_\Lambda = G_\Lambda\cdot A_\Lambda/\sqrt{S_\Lambda(x)S_\Lambda(y)} \to G$ in $L^2(M\times M)$ (the second factor $\to 1$ uniformly, and $\sup_\Lambda\|G_\Lambda\|_2 < \infty$). Multiplying C_Λ by the positive constant A_Λ leaves ranks unchanged, so $\mathrm{rank}\|N_\Lambda(X)-N_\Lambda(Y)\| = \mathrm{rank}(-H_\Lambda(X,Y))$. *(4) Rank convergence.* A bounded pair density carries $H_\Lambda \to G$ in L^2 to convergence in probability under the law of (X,Y); atomlessness of the limit makes the distribution functions converge, and the population Spearman/Kendall identities give $\rho_S(d_M, -H_\Lambda) \to \rho_S(d_M, -G)$. ☐

Corollary 26 (Uniform rank preservation on two-point homogeneous spaces). *If, in addition, M is compact two-point homogeneous, then $G(x,y) = g(d_M(x,y))$ with g strictly decreasing on $(0, \mathrm{diam}\, M)$; hence $\rho_G(M) = 1$ and $\rho_S\big(d_M(X,Y), \|N_\Lambda(X)-N_\Lambda(Y)\|\big) \to 1$, bounded below uniformly beyond a finite cutoff. This proves the eventual uniform-in-m rank form of Conjecture 3 on $S^2, S^3, \mathbb{RP}^2, \mathbb{RP}^3$ (the circle being the exact Fourier case above).*

Proof. Two-point homogeneity makes the invariant kernel a function of geodesic distance alone. With $A(r)$ the area density of a geodesic sphere, the zero-mean Green function satisfies $\Delta g = -1/\mathrm{vol}\, M$ off the pole, i.e. $A(r)\,g'(r) = -1 + \mathrm{vol}(M)^{-1}\int_0^r A(s)\,ds$; since $\int_0^r A < \mathrm{vol}\, M$ for $r < \mathrm{diam}\, M$, $g'(r) < 0$, so $-G$ is strictly increasing in d_M and the population rank correlation is 1. ☐

Graph transfer of the shrinking signal. The rank-carrying angular fluctuation has amplitude $\asymp A_\Lambda^{-1/2}$, so the growing-window transfer condition sharpens. With the calibration error $E_{n,m} = \max_i\|Q_n\widetilde{\Psi}_i - F_m(x_i)\|$ of Appendix B and $R_m \asymp \sqrt{A_m}$, normalization stability gives $\big|A_m\langle\widehat{q}_i,\widehat{q}_j\rangle - A_m\langle N_m(x_i), N_m(x_j)\rangle\big| \le C\,E_{n,m}\sqrt{A_m}$. The rank-transfer condition is therefore $E_{n,m}\sqrt{A_m} \to 0$—that is, $E_{n,m}\sqrt{\log\mu_m} \to 0$ $(d=2)$ or $E_{n,m}\,\mu_m^{1/4} \to 0$ $(d=3)$: graph calibration must converge *faster* than the rank-carrying signal shrinks, sharpening the bare $E_{n,m} \to 0$.

Conjecture 27 (Green-rank positivity). *For the stated geometric class and pair law, $\rho_G(M) = \rho_S\big(d_M(X,Y), -G_M(X,Y)\big) > 0$.*

This reorganizes the frontier. The *rank* component is proved: for $d \le 3$ the commute angular ranking converges uniformly in the cutoff to the Green-kernel ranking (Theorem 25), positive and equal to 1 on the circle, flat tori (with a computable coefficient), and two-point homogeneous spaces of dimension ≤ 3 (Corollary 26). What remains is the purely geometric question of Green-rank positivity, and the exact boundary is $d = 4$, where $\sum_k \mu_k^{-2} = \infty$ and the commute Green kernel ceases to be Hilbert–Schmidt (a stronger filter $\mu_k^{-s/2}$, $s > d/4$, restores the limit). In one line: uniform-in-mode *metric* control fails for the commute embedding, but uniform-in-mode *rank* control has a Green-kernel limit below the critical dimension $d = 4$, and Angular Preservation reduces to positivity of a geometric coefficient—settled on low-dimensional two-point homogeneous spaces.

Proof of Proposition 16 (concentration)

Plain filter. With $E_\Lambda(x,y) = \sum_{0<\mu_k\le\Lambda} \varphi_k(x)\varphi_k(y)$ (constant mode removed),

$$\left\|F_\Lambda^{(1)}(x) - F_\Lambda^{(1)}(y)\right\|^2 = E_\Lambda(x,x) + E_\Lambda(y,y) - 2E_\Lambda(x,y), \qquad R_\Lambda^{(1)}(x)^2 = E_\Lambda(x,x).$$

The pointwise Weyl law with uniform remainder [30] gives $E_\Lambda(x,x) = C_d\Lambda^{d/2} + O(\Lambda^{(d-1)/2})$ with spatially constant leading coefficient, and for $d_M(x,y) \ge \delta$ the off-diagonal spectral function obeys $|E_\Lambda(x,y)| = O(\Lambda^{(d-1)/2})$ uniformly, oscillating in the pair separation on the wavelength scale $\Lambda^{-1/2}$ (the Bessel-type scaling limit of the spectral projector [29]). The radial difference is $R_\Lambda^{(1)}(x) - R_\Lambda^{(1)}(y) = \big(E_\Lambda(x,x) - E_\Lambda(y,y)\big)/\big(R_\Lambda^{(1)}(x) + R_\Lambda^{(1)}(y)\big) = O(\Lambda^{(d-1)/2-d/4})$, so its square is $O(\Lambda^{d/2-1})$. Feeding these into the identity of Lemma 5,

$$\left\|N_\Lambda^{(1)}(x) - N_\Lambda^{(1)}(y)\right\|^2 = \frac{2C_d\Lambda^{d/2} + O(\Lambda^{(d-1)/2})}{C_d\Lambda^{d/2}\big(1 + O(\Lambda^{-1/2})\big)} = 2 + O\big(\Lambda^{-1/2}\big)$$

uniformly over separated pairs, with pair-dependence carried by $-2E_\Lambda(x,y)/E_\Lambda(x,x)$. The numerator kernel oscillates with sign changes on the wavelength scale, so it induces no monotone ordering in d_M. Concentration by itself, however, does not force the rank correlation to zero (a shrinking correction can preserve order); the exact collapse of both the Spearman and Kendall correlations against geodesic distance is established on the circle—where E_Λ is the Dirichlet kernel—in Theorem 17 below. This is the continuum mechanism behind the plain-eigenmap rank separation in the weighting ablation (§5.3).

Commute filter. The same computation with weights μ_k^{-1} replaces E_Λ by $G_\Lambda(x,y) = \sum_{0<\mu_k\le\Lambda} \mu_k^{-1}\varphi_k(x)\varphi_k(y)$: the diagonal is $C_d'\Lambda^{d/2-1}$ ($d > 2$; $C_2 \log\Lambda$, $d = 2$) with spatially constant leading coefficient (Step 7 of the Proposition 15 proof), the radial term is again lower order, and

$$\left\|N_\Lambda(x) - N_\Lambda(y)\right\|^2 = 2 - \frac{2\,G_\Lambda(x,y)}{G_\Lambda(x,x)}\big(1 + o(1)\big).$$

The correction kernel decays like μ^{-1}, and for $d \le 3$ (where $\sum_k \mu_k^{-2} < \infty$) it converges off the diagonal, in L^2, to the Green's function of M—a distance-structured kernel, and on a *two-point homogeneous* (isotropic) space a function of d_M alone. For $d \ge 4$ the sharp truncated Green kernel is no longer Hilbert–Schmidt stable and high-frequency oscillations can enter, so we claim no low-frequency dominance there. Both filters therefore concentrate at fixed scale as $\Lambda \to \infty$ (the wavelength story of Proposition 15), but the *ordering* information that rank statistics read is carried by a coherent kernel for commute and an oscillatory one for plain: the ablation's rank separation, in the continuum. □

Proof of Theorem 17 (exact plain-filter rank collapse on $\mathbb{S}^1$)

Write $\theta = x - y$; by evenness the chord depends only on the geodesic distance $|\theta| \in [0, \pi]$. Since $\|F_N\|^2 \equiv N$,

$$\langle \widehat{F}_N(x), \widehat{F}_N(y) \rangle = \frac{1}{N} \sum_{k=1}^{N} \cos(k\theta), \qquad Q_N(\theta) = 2 - \tfrac{2}{N} Z_N(\theta), \quad Z_N(\theta) := \sum_{k=1}^{N} \cos(k\theta).$$

The Dirichlet-kernel identity $1 + 2\sum_{k=1}^{N} \cos(k\theta) = \sin\big((N + \frac{1}{2})\theta\big)/\sin(\theta/2)$ gives, with $a(\theta) = 1/(2\sin(\theta/2))$ and $\omega_N = N + \frac{1}{2}$,

$$W_N(\theta) := Z_N(\theta) + \tfrac{1}{2} = a(\theta)\, \sin(\omega_N \theta).$$

As Q_N is a strictly decreasing affine function of Z_N, and Z_N, W_N differ by a constant, the three share the same rank correlations with Θ up to a global sign; it suffices to prove $\rho_S(\Theta, W_N(\Theta)) \to 0$ and $\tau_K(\Theta, W_N(\Theta)) \to 0$.

Step 1 (two-scale equidistribution). Fix $\varepsilon > 0$; on $[\varepsilon, \pi]$ the envelope a is continuous and bounded. For any $\varphi(\theta, t)$ bounded, continuous, and 2π-periodic in t,

$$\frac{1}{\pi} \int_{\varepsilon}^{\pi} \varphi(\theta, \omega_N \theta)\, d\theta \;\longrightarrow\; \frac{1}{\pi} \int_{\varepsilon}^{\pi} \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta, t)\, dt\, d\theta,$$

by expanding φ in Fourier modes in t: the zeroth mode is the phase average, and each nonzero mode is an oscillatory integral $\int_{\varepsilon}^{\pi} c_m(\theta) e^{im\omega_N\theta}\, d\theta \to 0$ by Riemann–Lebesgue as $\omega_N \to \infty$. Hence $\big(\Theta, W_N(\Theta)\big) \Rightarrow \big(\Theta,\, a(\Theta) \sin T\big)$ with $T \sim \mathrm{Unif}[0, 2\pi]$ independent of Θ (the removed collar $[0, \varepsilon)$ has probability $\varepsilon/\pi \to 0$, and $a(\Theta) < \infty$ a.s.). The limit $W := a(\Theta) \sin T$ is atomless and symmetric, so its CDF H is continuous with $H(-z) = 1 - H(z)$.

Step 2 (Spearman). For continuous variables the population Spearman coefficient is $\rho_S(U, V) = 12\, \mathbb{E}[F_U(U) F_V(V)] - 3$. With $F_\Theta(\Theta) = \Theta/\pi$ and H_N the CDF of $W_N(\Theta)$,

$$\rho_S(\Theta, W_N(\Theta)) = 12\, \mathbb{E}\Big[\tfrac{\Theta}{\pi}\, H_N\big(W_N(\Theta)\big)\Big] - 3.$$

Weak convergence to the atomless H gives $\sup_z |H_N(z) - H(z)| \to 0$ (Pólya), so H_N may be replaced by H. Step 1 then yields

$$\mathbb{E}\Big[\tfrac{\Theta}{\pi} H\big(W_N(\Theta)\big)\Big] \longrightarrow \frac{1}{\pi} \int_0^{\pi} \frac{\theta}{\pi} \Big(\frac{1}{2\pi} \int_0^{2\pi} H\big(a(\theta) \sin t\big)\, dt \Big) d\theta.$$

Pairing t with $t + \pi$ gives $H(a \sin t) + H(-a \sin t) = 1$, so the inner average is $\frac{1}{2}$ and the outer integral is $\frac{1}{4}$. Hence $\rho_S(\Theta, W_N(\Theta)) \to 12 \cdot \frac{1}{4} - 3 = 0$, and by the affine reversal $\rho_S(\Theta, Q_N(\Theta)) \to 0$.

Step 3 (Kendall). Take independent copies (Θ_1, Θ_2) and write $\tau_K = \mathbb{E}\big[\mathrm{sgn}(\Theta_1 - \Theta_2)\, \mathrm{sgn}(W_N(\Theta_1) - W_N(\Theta_2))\big]$. Applying the Fourier/Riemann–Lebesgue argument of Step 1 jointly, the four-tuple $\big(\Theta_1, \Theta_2, \omega_N\Theta_1, \omega_N\Theta_2\big)$ converges weakly to $(\Theta_1, \Theta_2, T_1, T_2)$ with $T_1, T_2 \sim \mathrm{Unif}[0, 2\pi]$ independent of one another and of (Θ_1, Θ_2) (each nonzero joint Fourier mode is killed by integrating first in whichever variable carries nonzero frequency). The integrand $\mathrm{sgn}(\theta_1 - \theta_2)\, \mathrm{sgn}(a(\theta_1) \sin t_1 - a(\theta_2) \sin t_2)$ is bounded and its discontinuity set $\{\theta_1 = \theta_2\} \cup \{a(\theta_1) \sin t_1 = a(\theta_2) \sin t_2\}$ is null under the limit law, so the portmanteau theorem gives $\tau_K \to \mathbb{E}\big[\mathrm{sgn}(\Theta_1 - \Theta_2)\, \mathrm{sgn}(a(\Theta_1) \sin T_1 - a(\Theta_2) \sin T_2)\big]$. Conditioned on (Θ_1, Θ_2) the inner sign is that of a difference of two independent atomless symmetric variables, so its expectation is 0; hence $\tau_K(\Theta, W_N(\Theta)) \to 0$ and $\tau_K(\Theta, Q_N(\Theta)) \to 0$. $\square$

Graph transfer at growing m, and the C^1 route

For fixed m, Appendix B is the complete transfer. For a growing spectral window $m = m(n)$, the same calibration goes through—with every constant carrying an m-subscript—whenever

$$\mathcal{E}_n(\mu_m)\,\gamma_m^{-1} \;\longrightarrow\; 0,$$

where γ_m is the spectral gap isolating the retained block and $\mathcal{E}_n(\mu_m)$ is the graph-Laplacian consistency error on continuum eigenfunctions up to frequency μ_m; since high-frequency eigenfunctions have polynomially large derivatives, $\mathcal{E}_n(\mu_m)$ grows polynomially in μ_m, so $m(n)$ may grow, but not too fast. The resulting cutoff is $\theta_{n,m} \gtrsim E_{n,m}/(a_m\rho_m^-)$, every factor now m-dependent. Separately, the additive pairwise error of the sup-norm argument is what forces θ_n to sit at the spectral-rate scale; a $C^{0,1}$ eigenvector theory—the Lipschitz estimates of [7] are the available ingredient—would replace it by a multiplicative derivative-level error ($\|d\hat{Z}(v)\| \geq (1-\eta)\|dN(v)\|$ once the C^1 calibration error is below $\eta\, a_m\rho_m^-$), pushing the cutoff down to the connectivity scale ε_n—optimal, since below it combinatorial twins exist (Appendix B). We state these as the conditions a growing-window theorem needs, not as a theorem.

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